

Received 23 May 2026, accepted 3 June 2026, date of publication 9 June 2026, date of current version 16 June 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3701422

RESEARCH ARTICLE

A Physically Interpretable Hybrid Framework for Spatial Modeling of Crowd-Modified Wet-Bulb Heat Risk During Mass Gatherings

FAYÇAL HAMDI¹, MOHAMED BENAIDA¹, HACHEMI BENNACEUR², YASSER ALDWYAN¹, HANI ALMOAMARI¹, ABDULAZIZ ALSHAHRANI¹, AND ABDULLAH ALSHANQITI¹

¹Faculty of Computer and Information Systems, Islamic University of Madinah, Madinah 42351, Saudi Arabia

²College of Computer and Information Sciences, Al Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh 11564, Saudi Arabia

Corresponding author: Fayçal Hamdi (fhamdi@iu.edu.sa)

ABSTRACT Extreme heat poses a critical risk in high-density mass gatherings, where atmospheric conditions, urban morphology, and crowd dynamics interact to produce hazardous microclimates. During the Hajj pilgrimage in Makkah, these effects are amplified by dense crowds, heat-retaining surfaces, and constrained ventilation. Conventional Urban Heat Island (UHI) indicators based on land surface temperature do not directly capture human thermophysiological stress and are therefore inadequate for operational heat-risk assessment. This paper introduces a physics-guided hybrid framework for modeling crowd-modified wet-bulb heat risk. The approach combines (i) wet-bulb temperature as a physiologically grounded baseline; (ii) a data-driven spatial amplification term derived from satellite-based morphological features; (iii) scenario-based estimates of crowd-induced heat contributions; and (iv) wind-related modulation that accounts for stagnation and dust effects. By explicitly separating time-varying atmospheric forcing from spatial susceptibility, the framework provides an interpretable decomposition of heat-risk drivers. The machine-learning model for estimating the spatial amplification component achieves a mean absolute error of 1.13 °C ($R^2 = 0.586$). Application of the framework to the 2021–2025 Hajj seasons suggests distinct site-specific heat-risk regimes, with Arafat and Muzdalifah dominated by urban amplification and Haram and Mina more strongly influenced by crowd density. Across all years, the combined influence of urban morphology, crowd density, and wind conditions results in an average increase of approximately 2.1 °C to 2.3 °C in effective wet-bulb temperature across sites. The 2024 Hajj season is identified as the most severe, with a substantial increase in extreme-risk exposure. The proposed framework provides a spatially explicit, physically interpretable, and decision-support-oriented methodology for heat-risk assessment within the Hajj context. The decomposition-based framework is, in principle, transferable to other large-scale mass-gathering environments, subject to retraining of the spatial amplification model and replacement of the event-specific crowd allocation component with context-appropriate crowd estimation approaches.

INDEX TERMS Crowd dynamics, environmental monitoring, heat stress, mass gatherings, physics-guided hybrid modeling, spatial modeling, urban heat island, wet-bulb temperature.

I. INTRODUCTION

Extreme heat is among the most dangerous environmental hazards in mass gatherings, where dense crowds, limited shade, and prolonged outdoor exposure converge to create

potentially lethal conditions [1], [2]. The Hajj pilgrimage in Makkah, Saudi Arabia, exemplifies this risk, gathering around 1.5–1.8 million participants annually, often during summer conditions when air temperatures exceed 45 °C [3], [4]. The 2024 Hajj season tragically illustrated these dangers, with over 1,300 heat-related fatalities reported by Saudi authorities amid temperatures reaching up to

The associate editor coordinating the review of this manuscript and approving it for publication was Larbi Boubchir¹.

51.8 °C [5], [6]. More than 80% of these deaths occurred among unauthorized pilgrims walking long distances without adequate shelter or cooling [7]. These outcomes, despite extensive mitigation measures and decades of operational experience, highlight the need for more effective, spatially explicit, and physiologically grounded heat-risk assessment methods for high-density religious mass gatherings [8], [9].

Understanding why such fatalities occur requires examining how the human body responds to extreme heat. As ambient temperature approaches skin temperature (approximately 35 °C), the body's ability to dissipate heat via convection and radiation sharply diminishes, making evaporative cooling via sweating the dominant thermoregulatory pathway. This mechanism fails when humidity is high, a condition conveniently quantified by wet-bulb temperature, which approximates the lowest temperature attainable by evaporative cooling under given atmospheric conditions [10], [11]. Wet-bulb temperatures above 35 °C are often cited as a theoretical limit for human survivability. However, recent studies demonstrate that critical physiological strain may occur at much lower levels, especially for vulnerable individuals performing physical activity in hot, radiant, and low-wind environments [12], [13]. In practice, wet-bulb temperatures in the range 28 °C to 32 °C already pose severe risks, particularly for vulnerable individuals and those engaged in prolonged exertion in crowded environments. This makes wet-bulb temperature a physiologically grounded metric that captures the combined effects of temperature and humidity on human cooling capacity [10].

Heat risk in this context is driven by multiple interacting factors that operate across scales:

- **Ambient meteorological conditions**, including air temperature, humidity, wind speed, and solar radiation, which establish the baseline thermal environment and determine the fundamental potential for evaporative and convective cooling [14].
- **Urban morphology and surface heat retention**, where building materials, geometry, and thermal properties modulate local energy balances, creating spatial heterogeneity in heat exposure through shading, multiple reflections, and nocturnal heat storage [15], [16], [17].
- **High-density crowds** that reduce airflow, impede evaporative cooling, and generate additional metabolic heat, thereby altering local humidity and effective temperature, particularly in semi-enclosed or tented spaces [9].
- **Nocturnal heat release from built infrastructure**, which limits overnight recovery, sustains elevated minimum temperatures, and enables cumulative heat stress across multiple days [15], [18].

These factors do not act independently; rather, they interact nonlinearly to produce emergent thermal conditions that exceed the sum of their individual contributions. Dense urban fabric can amplify daytime heat absorption while restricting ventilation. In contrast, high-albedo or shaded areas can

partially offset radiative loads [17]. Crowds further modify the local environment by generating metabolic heat, increasing humidity, and reducing airflow [9]. In occupied Hajj tents, relative humidity has been observed to rise by 15–28 percentage points above outdoor levels. Pilgrims experience severe thermal discomfort and physiological strain for most of their time inside. This illustrates how crowd presence can transform the microclimate independently of background meteorology [19]. Characterizing heat risk therefore requires accounting for both environmental thermodynamics and human physiology within a spatially and temporally explicit framework that captures these interacting processes.

The spatial heterogeneity of these factors during Hajj creates distinct microclimatic zones across the pilgrimage sites. The plain of Arafat, where pilgrims spend the afternoon in largely open worship, experiences different thermal dynamics than the dense tent city of Mina or the multi-level Jamarat Bridge structure, each with unique combinations of sky-view factor, surface albedo, anthropogenic heat, and ventilation pathways [9], [16]. Empirical and modeling studies of Hajj logistics and crowd safety have highlighted that risk is strongly location- and time-dependent, with particular sensitivity around the peak afternoon hours in Arafat and during congested movements near Jamarat [2]. Moreover, these conditions vary throughout the day and across the multi-day pilgrimage, with cumulative exposure and limited nighttime relief emerging as critical determinants of health outcomes in a warming climate [15], [18].

Urban Heat Island (UHI) studies have traditionally quantified temperature differences between urban and rural areas using land surface temperature (LST) metrics derived from thermal satellite imagery, following the foundational urban climate framework established by Oke [15]. LST-based indicators are valuable for understanding urban thermal structure and identifying macro-scale hotspots. However, surface temperature does not directly determine human thermoregulatory stress. Human heat response depends primarily on near-surface air temperature, humidity, wind, and radiation, as captured by metrics such as wet-bulb temperature, heat index, WBGT, and the Universal Thermal Climate Index [14], [20], [21]. Recent work has further emphasized that satellite-derived LST can diverge substantially from canopy-layer air temperature, especially during daytime, and may lack the spatiotemporal specificity required for actionable municipal heat-mitigation planning at the neighborhood scale [20], [22]. Moreover, conventional UHI approaches typically treat urban areas as static thermal anomalies and do not explicitly represent dynamic modifications introduced by transient crowd presence. These include metabolic heat generation, humidity buildup, and wind shielding. Such limitations become particularly consequential in mass-gathering contexts, where the crowd itself becomes a dominant modifier of the thermal environment [9], [23].

Parallel to these developments, there has been growing interest in integrating meteorological, urban, and human factors into more holistic heat-risk frameworks. Smart-city

digital-twin approaches, for example, have combined urban sensing, forecasting, and crowd-aware exposure estimation to assess collective heat exposure during large urban events [23]. In the Hajj context, pilot systems based on Internet of Things (IoT) wearables and environmental sensing have been proposed to deliver real-time, personalized heat-stress alerts and clinical escalation pathways for high-risk pilgrims. However, these systems remain largely individual-focused and are not yet coupled with spatially explicit crowd-modified microclimate models [24]. More broadly, recent reviews of machine learning for urban heat highlight a shift from temperature-based models toward approaches that incorporate thermal comfort and population exposure. Despite this progress, most studies still neglect explicit crowd thermodynamics and focus on residential or city-wide settings rather than short-duration, high-density mass gatherings [21].

Recent advances in hybrid physically guided machine learning and physics-aware modeling offer new opportunities to address these challenges by combining physical understanding with data-driven spatial modeling. Physics-informed neural networks and related approaches embed conservation laws, constitutive relationships, or mechanistic constraints into flexible statistical models, improving generalization, robustness, and interpretability for complex spatiotemporal processes [25], [26]. In the context of urban heat, machine learning models have been used to downscale reanalysis products, assess the impact of urban form and mitigation strategies on microclimate, and project future heat burdens under climate-change scenarios [27].

However, most existing physically guided or hybrid machine-learning approaches for heat-risk assessment either (i) focus on air or surface temperature rather than physiologically grounded metrics such as wet-bulb temperature, or (ii) provide spatially detailed, data-driven estimates without explicitly representing crowd-induced modifications or clearly separating time-varying atmospheric forcing from fixed spatial susceptibility [21].

To the best of our knowledge, no existing study jointly models crowd-modified wet-bulb heat risk across the main Hajj sites within a physics-guided hybrid modeling framework, nor does it provide a systematic multi-year comparative analysis across Hajj seasons.

The key novelty of this work lies in explicitly integrating crowd-induced thermodynamic effects into a physiologically grounded heat-risk framework based on wet-bulb temperature. Unlike approaches that treat urban heat or human exposure in isolation, the proposed model provides a unified and interpretable decomposition separating atmospheric forcing, urban morphological amplification, and crowd-driven microclimate modifications. This enables spatially explicit characterization of heat-risk drivers in large-scale mass gatherings, a setting that remains comparatively underexplored in prior heat-risk modeling studies.

Building on these insights, we propose a unified physics-guided hybrid framework for spatial modeling of

crowd-modified wet-bulb heat risk during Hajj. The approach combines meteorological wet-bulb temperature as a baseline physiological hazard with data-driven estimates of urban amplification derived from satellite-based morphological features, while explicitly incorporating wind modulation and crowd-induced heat contributions. By separating time-varying atmospheric forcing from fixed spatial susceptibility, the model provides an interpretable decomposition of effective heat risk into physically interpretable components while preserving interpretable and physically bounded component contributions through constrained additive parameterization.

The main contributions of this work are summarized as follows:

- A physics-guided hybrid modeling framework for spatial heat-risk assessment in large-scale mass gatherings, integrating wet-bulb temperature with data-driven representations of urban morphology and scenario-based crowd-induced thermal effects under plausible crowd distributions.
- An interpretable additive decomposition of effective wet-bulb temperature into meteorological baseline, urban amplification, crowd metabolic heat contributions, and wind-modulation effects.
- A spatial modeling approach that separates time-varying atmospheric forcing from fixed spatial susceptibility using satellite-derived morphological features.
- A comprehensive multi-year analysis across five Hajj seasons (2021–2025), revealing consistent spatial patterns and site-specific heat-risk regimes under varying climatic conditions.
- A prototype decision-support platform for scenario-based heat-risk monitoring across the principal Hajj pilgrimage sites.

The effectiveness of the proposed framework is demonstrated using Hajj-season data from 2021 to 2025, covering a wide range of climatic conditions and crowd densities. The machine-learning component reproduces persistent spatial variations in urban heat amplification from satellite-derived morphological features, while the full hybrid framework suggests distinct site-specific heat-risk regimes across the pilgrimage network. In particular, Arafat and Muzdalifah emerge as locations with strong intrinsic amplification driven by surface thermal properties and heat storage, whereas Haram and Mina are more strongly influenced by crowd-induced microclimatic effects under the modeled crowd distributions.

Across all years, the combined influence of urban morphology, crowd density, and wind conditions produces a consistent increase of approximately 2.1 °C to 2.3 °C in effective wet-bulb temperature relative to the meteorological baseline. The 2024 Hajj season stands out as the most severe, characterized by the alignment of elevated atmospheric forcing and strong spatial amplification, resulting in a substantial increase in extreme-risk (BLACK-alert) exposure, particularly at Arafat and Muzdalifah. These results suggest

that extreme heat risk during large-scale mass gatherings is not driven by meteorological conditions alone, but also by the combined influence of environmental and crowd-related amplification mechanisms.

The remainder of this paper is organized as follows. Section II presents the mathematical formulation of the hybrid heat-risk framework. Section III describes the datasets and experimental design. Section IV introduces the machine learning model for estimating spatial amplification. Section V presents the results, including spatial patterns, component analysis, sensitivity assessment, and multi-year comparisons. Section VI describes the prototype decision-support system for heat-risk monitoring. Section VII discusses the interpretation of results, uncertainty, and implications for heat-risk management. Section VIII presents the open-source implementation and reproducibility resources. Finally, Section IX concludes the paper and outlines future directions.

II. PROPOSED FRAMEWORK

This section presents the hybrid physics-guided framework for modeling crowd-modified wet-bulb heat risk. We first formulate the problem as an additive decomposition of heat risk components, then describe each component in detail, and finally explain how machine learning is used to estimate spatial amplification potential from satellite-derived morphological features.

A. PROBLEM FORMULATION

Let $x \in \Omega \subset \mathbb{R}^2$ denote spatial location within the Hajj pilgrimage sites (Mina, Arafat, Muzdalifah, and the Haram area), and let t denote time during the Hajj season. Our objective is to estimate a spatially explicit heat-risk field

$$H(x, t)$$

that represents physiologically relevant thermal stress experienced by pilgrims, expressed here as an effective wet-bulb temperature (used as a proxy for heat risk).

Rather than treating heat risk as a monolithic variable, the proposed framework decomposes it into interpretable physical components that represent first-order amplification mechanisms. Specifically, we separate a baseline atmospheric forcing term from a set of additive modifiers associated with urban morphology, crowd concentration, and wind conditions:

$$H(x, t) = H_0(t) + \Delta T_{urban}(x, t) + \Delta T_{crowd}(x, t) + \Delta T_{wind}(t)$$

where:

- $H_0(t)$ denotes the baseline atmospheric thermal stress derived from meteorological conditions (e.g., wet-bulb temperature), representing the large-scale environmental forcing that affects all locations simultaneously.
- $\Delta T_{urban}(x, t)$ represents spatial amplification caused by urban morphology and heat storage in the built environment, reflecting the tendency of dense urban surfaces to retain and release heat.

- $\Delta T_{crowd}(x, t)$ captures the additional thermal load generated by dense crowds through metabolic heat emission and reduced ventilation within highly populated areas.
- $\Delta T_{wind}(t)$ represents wind-related modulation of thermal stress, accounting for stagnant-air effects under low wind conditions and additional environmental stress associated with dust entrainment during stronger winds.

This decomposition reflects the hybrid structure of the proposed model. Meteorological forcing varies primarily in time, spatial susceptibility is governed by urban morphology, and event-driven modifiers such as crowd density and wind conditions introduce additional temporal variability. The additive formulation preserves interpretability by allowing each physical process to be examined independently while maintaining a transparent estimate of total heat stress.

The following subsections detail the formulation of each component.

B. BASELINE PHYSIOLOGICAL HEAT STRESS

A robust heat risk assessment must be grounded in human physiology. While air temperature is commonly reported in weather forecasts, it does not adequately capture the body's ability to cool itself through sweating. This cooling capacity is fundamentally limited by humidity: when the air is already saturated with moisture, sweat cannot evaporate, and the body overheats.

Human thermal strain is therefore strongly correlated with wet-bulb temperature T_w , which reflects the combined effects of temperature and humidity on evaporative cooling. Wet-bulb temperature represents the lowest temperature that can be achieved through evaporative cooling and is directly related to the physiological wet-bulb limit for human survivability.

Given air temperature $T_a(t)$ and dew point temperature $T_d(t)$ from hourly station observations, we first compute vapor pressure using the Magnus formula:

$$e(T) = 6.112 \exp\left(\frac{17.67 T}{T + 243.5}\right)$$

where T is in degrees Celsius and e is in hectopascals. Relative humidity is then derived as:

$$RH = 100 \cdot \frac{e(T_d)}{e(T_a)}$$

With temperature and relative humidity known, wet-bulb temperature is approximated using the Stull [28] formula, which provides a robust analytical approximation suitable for operational deployment:

$$\begin{aligned} T_w = & T_a \arctan\left(0.151977\sqrt{RH} + 8.313659\right) \\ & + \arctan(T_a + RH) - \arctan(RH - 1.676331) \\ & + 0.00391838 RH^{1.5} \times \arctan(0.023101 RH) \\ & - 4.686035 \end{aligned}$$

This formulation has been validated across a wide range of atmospheric conditions and offers computational efficiency

for large-scale spatial modeling. The baseline heat stress is therefore defined as:

$$H_0(t) = T_w(t)$$

This term captures atmospheric physiological stress independent of spatial morphology, serving as the foundation upon which local amplification effects are added.

C. URBAN MORPHOLOGICAL AMPLIFICATION

While the baseline hazard $H_0(t)$ captures regional atmospheric forcing, the actual heat exposure experienced at a given location also depends on local urban morphology and surface thermal properties. Built materials such as concrete and asphalt absorb solar radiation during the day and release the stored energy gradually after sunset, producing spatially heterogeneous heat amplification commonly associated with the urban heat island effect.

To represent this process, we introduce a location-dependent amplification potential $A(x)$ that quantifies the persistent tendency of each location to elevate thermal stress relative to the meteorological baseline. This spatial susceptibility reflects the combined influence of urban form, surface materials, vegetation cover, and thermal inertia. The estimation of $A(x)$ from satellite-derived morphological features is described later in Section IV.

The urban contribution to effective heat stress is modeled as a separable product of spatial susceptibility and diurnal modulation:

$$\Delta T_{urban}(x, t) = A(x) \cdot \Phi(h)$$

where:

- $A(x)$ denotes the spatial amplification potential at location x ,
- $\Phi(h)$ is a diurnal modulation factor dependent on hour of day h ,
- the multiplicative structure ensures that locations with greater susceptibility experience proportionally stronger amplification during periods of peak heat release.

The diurnal modulation function accounts for the thermal inertia of urban materials, which causes maximum heat release to occur during the evening rather than at the time of maximum solar input. Based on empirical observations of urban heat island dynamics, this modulation is represented using a cosine function:

$$\Phi(h) = 1 + \gamma \cos\left(\frac{2\pi(h - h_0)}{24}\right)$$

where $\gamma = 0.18$ sets the modulation amplitude (range 0.82–1.18) and $h_0 = 20$ (8 PM) denotes the hour of peak release of stored heat. This formulation captures the delayed emission of energy from urban surfaces. This phenomenon limits nocturnal cooling and can promote the accumulation of thermal stress across successive days. The chosen value of γ yields a $\pm 18\%$ diurnal modulation around the mean, consistent with reported day–night contrasts and

evening–peaking canopy-layer and surface urban heat island intensities in arid and semi-arid cities [29].

The amplification potential is constrained within physically plausible bounds based on observed magnitudes of urban heat island intensity in arid environments:

$$0 \leq A(x) \leq \Delta T_{max}$$

where $\Delta T_{max} = 5.0^\circ\text{C}$ represents a conservative upper bound consistent with reported surface–air temperature differences in dense Middle Eastern cities [30]. This bound ensures that the modeled urban amplification remains physically realistic even under extreme morphological conditions.

D. CROWD METABOLIC CONTRIBUTION

Mass gatherings create a distinct thermal environment compared with conventional urban heat exposure because large numbers of people concentrated in confined areas release metabolic heat and modify airflow patterns. During the Hajj pilgrimage, dense pedestrian concentrations can locally elevate perceived heat stress even when background meteorological conditions remain unchanged.

Let $\rho_n(x, t) \in [0, 1]$ denote the normalized crowd density at location x and time t , derived from site capacity utilization and crowd distribution estimates. The direct thermal contribution of the crowd is modeled as a bounded additive offset:

$$\Delta T_{crowd}(x, t) = \kappa \rho_n(x, t)$$

where κ represents the maximum crowd-related thermal amplification. We set $\kappa = 1.2^\circ\text{C}$, corresponding to a modest, order-of-magnitude consistent temperature increase at extreme crowd densities and aligning with reports that natural climatic heat load during Hajj is further potentiated by micro-scale heating from pilgrim crowds and vehicular activity [31].

This formulation assumes that the local thermal contribution of human metabolism scales approximately linearly with crowd density. The resulting term is bounded:

$$0 \leq \Delta T_{crowd}(x, t) \leq \kappa$$

Wind effects are treated separately through the convective modulation component described in the following subsection. This separation simplifies the model structure and improves interpretability by distinguishing metabolic heat generation from atmospheric transport processes.

E. WIND CONVECTIVE MODULATION

Wind plays a fundamental role in human thermal exposure by regulating convective heat exchange between the body and the surrounding environment. Moderate airflow promotes heat removal and improves ventilation, while stagnant conditions suppress convective transport and allow heat and humidity to accumulate near the surface. In arid environments, strong winds may also entrain dust particles that introduce an additional environmental burden.

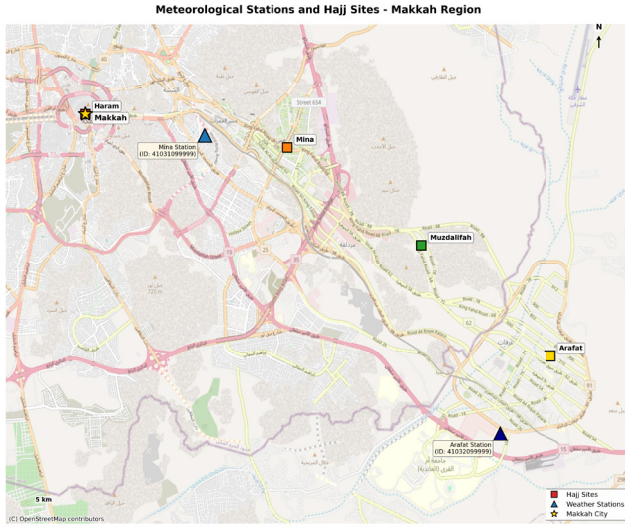


FIGURE 1. Spatial distribution of meteorological stations (Mina and Arafat) relative to the four Hajj sites.

To represent these effects, wind influence is modeled through two bounded additive contributions corresponding to stagnant-air conditions and dust entrainment.

1) STAGNATION EFFECT

Very low wind speeds reduce atmospheric mixing and inhibit ventilation. This stagnation effect is represented by a smooth logistic function that decreases as wind speed increases:

$$\Delta T_{stagnant}(t) = \frac{\alpha_s}{1 + \exp\left(\frac{w(t) - w_s}{\beta_s}\right)}$$

where $w(t)$ is the wind speed at time t , α_s denotes the maximum stagnation penalty under near-zero wind conditions, w_s represents the transition wind speed at which ventilation begins to improve, and β_s controls the smoothness of the transition.

2) DUST ENTRAINMENT

At higher wind speeds, airborne dust may increase environmental stress by reducing air quality and partially offsetting the benefits of ventilation. This effect is modeled using a second logistic function:

$$\Delta T_{dust}(t) = \frac{\alpha_d}{1 + \exp\left(-\frac{w(t) - w_d}{\beta_d}\right)}$$

where α_d defines the maximum dust-related penalty, w_d represents the wind speed at which dust entrainment becomes significant, and β_d controls the transition width.

The overall wind-related contribution is therefore expressed as the sum of the stagnation and dust effects:

$$\Delta T_{wind}(t) = \Delta T_{stagnant}(t) + \Delta T_{dust}(t)$$

This formulation ensures physically plausible behavior across the full range of wind conditions. When wind speeds

are very low, $\Delta T_{stagnant}$ dominates due to poor ventilation. As wind speed increases, stagnation effects diminish while the dust term may increase gradually. Both contributions remain bounded, preventing unrealistic amplification of thermal stress.

III. DATASETS AND EXPERIMENTAL DESIGN

The hybrid heat-risk framework developed in this study integrates multiple data sources describing both atmospheric forcing and urban morphological characteristics. Temporal inputs are derived from meteorological observations and scenario-based crowd distribution estimates, while spatial susceptibility features are constructed from satellite-based measurements of urban form and surface properties. Together, these datasets enable the model to represent both time-varying physiological heat stress and the persistent spatial amplification of heat associated with urban morphology.

In the following subsections, we describe the meteorological datasets, satellite-derived spatial features, and crowd estimation methodology used to construct the model inputs. We then outline the experimental protocol employed for model training, evaluation, and sensitivity analysis.

A. METEOROLOGICAL DATA

Hourly meteorological observations were obtained from automated weather stations located at Mina and Arafat for the period 2021–2025, covering five Hajj seasons, each with varying climatic conditions and pilgrim densities. The data were retrieved from the publicly available KAPSARC Data Portal [32], which provides integrated surface climate observations for Saudi Arabia.

For this study, records were extracted from the Mina station (station ID: 41031099999) and the Arafat station (station ID: 41032099999), which are the closest meteorological stations to the main Hajj ritual sites. Following the temporal definition adopted in the crowd estimation model, the dataset was restricted to the Hajj window spanning from day -2 to day $+5$, where day 0 corresponds to 8 Dhul-Hijjah (arrival in Mina). This filtering yields a focused subset of 625 hourly records (310 from Mina and 315 from Arafat) representing the eight-day Hajj window in each year.

Figure 1 shows the spatial distribution of the two meteorological stations relative to the four principal Hajj sites.

The following variables were extracted from the hourly observations:

- Air temperature (T_a , °C)
- Dew point temperature (T_d , °C)
- Wind speed (W , m s^{-1})
- Atmospheric pressure (hPa)
- Timestamp (Asia/Riyadh timezone)

The source dataset includes quality-control flags provided by the Saudi National Center for Meteorology (NCM) and archived through NOAA's National Centers for

TABLE 1. Summary statistics for meteorological variables at Mina and Arafat stations during Hajj seasons (2021–2025).

Station	Variable	N	Mean	Std	Min	Q1	Median	Q3	Max
Mina	Air Temperature (°C)	310	36.0	3.9	28.0	33.2	35.4	38.8	45.8
Mina	Dew Point (°C)	310	13.3	5.8	-3.0	9.0	13.0	17.0	26.0
Mina	Relative Humidity (%)	310	28.3	12.7	5.0	18.4	25.9	35.9	77.0
Mina	Wet-bulb Temperature (°C)	310	21.9	2.9	15.5	19.8	21.7	23.9	29.8
Mina	Wind Speed (m/s)	310	2.1	1.5	0.0	1.0	2.1	3.1	7.7
Arafat	Air Temperature (°C)	315	36.9	4.6	28.0	33.4	36.4	40.4	48.4
Arafat	Dew Point (°C)	315	13.0	6.3	-6.0	10.0	13.0	17.0	26.0
Arafat	Relative Humidity (%)	315	26.8	12.6	4.4	17.9	25.5	33.1	73.1
Arafat	Wet-bulb Temperature (°C)	315	22.0	3.0	14.4	20.1	21.7	24.2	30.5
Arafat	Wind Speed (m/s)	315	1.8	1.8	0.0	0.0	2.1	3.1	6.7

Environmental Information (NCEI), ensuring reliable observational records.

Table 1 summarizes the distribution of key meteorological variables across the 2021–2025 Hajj seasons.

These observations form the basis for estimating the baseline physiological heat stress. Relative humidity (RH) is derived from air temperature and dew point temperature using the Magnus formulation, while wet-bulb temperature (T_{wb}) is computed using the analytical approximation proposed by Stull [28]. The resulting T_{wb} time series represents the baseline hazard $H_0(t)$ in the heat-risk model.

It should be noted that meteorological forcing for the Haram and Muzdalifah sites is extrapolated from the Mina and Arafat stations, as dedicated meteorological stations are not available at these locations. Haram is situated in a dense urban canyon environment where wind speeds may be systematically lower and temperatures may differ from the exposed station locations, while Muzdalifah is open terrain that may experience different radiative cooling and wind exposure. The wind-related terms (ΔT_{wind}) are applied uniformly across all sites in the current implementation. This spatial extrapolation introduces uncertainty, particularly for site-specific wind-driven effects at Haram and Muzdalifah.

B. SATELLITE-DERIVED SPATIAL FEATURES

To characterize spatial variations in urban heat amplification, we constructed a set of satellite-derived morphological features on a regular 1 km $\times$ 1 km grid covering the Makkah study area. The grid was generated in UTM Zone 37N (EPSG:32637) to ensure metric consistency across spatial datasets, with each grid cell assigned a unique identifier and geographic centroid.

1) GRID CONSTRUCTION AND HARMONIZATION

All spatial layers were reprojected to the common UTM 37N coordinate system and resampled to align with the reference grid. Continuous raster datasets were aggregated using area-weighted averaging, while categorical layers (e.g., Local Climate Zones) were resampled using mode selection to preserve the dominant class within each cell. This harmonization ensures that each grid cell contains a consistent set of features derived from multiple satellite platforms and geospatial sources.

2) URBAN MORPHOLOGICAL FEATURES

Urban morphology was characterized using two complementary datasets.

a: BUILDING DENSITY

Building footprints were obtained from the Microsoft global building footprint dataset. For each grid cell, the footprint polygons were clipped to the cell boundary and aggregated to compute total built-up area. Building density was then calculated as

$$BD(x) = \frac{\text{built-up area}}{\text{cell area}}$$

yielding values in the range $[0, 1]$ that represent the intensity of urban development.

b: LOCAL CLIMATE ZONES (LCZ)

Urban form classification was derived from the World Urban Database and Portal Tools (WUDAPT) Local Climate Zone product. LCZ categorizes urban areas into 17 classes based on building height, density, and surface materials. The LCZ raster was resampled to the reference grid using categorical mode selection so that each grid cell receives its dominant morphological class.

Vegetation cover was characterized using Sentinel-2 derived Normalized Difference Vegetation Index (NDVI) composites. Mean growing-season NDVI was computed for each grid cell, providing a continuous indicator of vegetation density $NDVI(x) \in [-1, 1]$ that reflects the cooling effects of evapotranspiration and shading.

3) SURFACE TEMPERATURE FEATURES

Surface thermal characteristics were derived from MODIS land surface temperature products. We obtained 8-day composites from both Terra (morning overpass) and Aqua (afternoon overpass) satellites (MOD11A2 and MYD11A2) at 1 km resolution.

For each grid cell we computed multi-year mean values $LST_{Terra}(x)$ and $LST_{Aqua}(x)$ from the Terra and Aqua observations respectively. The mean land surface temperature used in subsequent analysis is defined as

$$LST_{mean}(x) = \frac{LST_{Terra}(x) + LST_{Aqua}(x)}{2}$$

TABLE 2. Summary of satellite-derived features.

Feature	Source	Resolution	Role in Model
LCZ Class	WUDAPT	100 m → 1 km	Categorical urban form predictor
Building Density	Microsoft Footprints	Vector → 1 km	Development intensity
NDVI	Sentinel-2	10 m → 1 km	Vegetation cooling effects
LST Terra/Aqua	MODIS	1 km	Surface temperature and thermal inertia
ERA5 Temperature	ERA5-Land	9 km → 1 km	Atmospheric baseline
Surface Excess	Derived (MODIS-ERA5)	1 km	ML regression target
Grid Summary	Generated	1 km	1,890 grid cells total

The diurnal surface temperature range

$$\Delta LST(x) = LST_{Aqua}(x) - LST_{Terra}(x)$$

serves as a proxy for thermal inertia, where smaller ranges indicate surfaces with greater heat storage capacity.

To represent the regional atmospheric background, we used the ERA5-Land reanalysis dataset provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5-Land provides hourly estimates of near-surface meteorological variables at approximately 0.1° spatial resolution (~ 9 km). In this study, the 2-m air temperature variable was extracted for the Hajj seasons of 2021–2025 and converted from Kelvin to degrees Celsius. Hourly observations were aggregated to compute multi-year mean air temperature fields representing the large-scale climatic baseline. The resulting variable $T_{ERA5,mean}(x)$ provides a regional atmospheric reference against which local surface temperatures derived from satellite observations can be compared.

4) SURFACE EXCESS TARGET

The target variable used to learn spatial amplification potential is the climatological surface excess $SE(x)$, defined as

$$SE(x) = LST_{mean}(x) - T_{ERA5,mean}(x)$$

where $LST_{mean}(x)$ is defined as the mean of Terra and Aqua observations as described above. This quantity captures the persistent difference between surface temperature and atmospheric temperature attributable to local urban morphology.

Table 2 summarizes all satellite-derived features, their sources, spatial resolutions, and roles in the model. The resulting dataset contains 1,890 grid cells covering the study domain, each associated with a complete feature vector used for training the spatial amplification model.

C. CROWD DISTRIBUTION ESTIMATION

Unlike meteorological and satellite datasets, which can be directly observed, hourly crowd distributions across Hajj sites are not systematically measured due to the absence of comprehensive sensor coverage. To construct the crowd features required by the heat-risk model, we therefore developed a synthetic but physically constrained crowd distribution model. The resulting crowd distributions are

therefore intended to represent plausible operational scenarios constrained by ritual logistics and site capacities rather than empirically validated reconstructions of true pilgrim movement patterns. The model combines four sources of information: (i) annual pilgrim totals reported by official statistics, (ii) the ritual schedule governing site occupancy during the Hajj pilgrimage, (iii) astronomical timings that determine transitions between ritual stages, and (iv) physical constraints on the rate at which pilgrims can move between sites.

A detailed description of the allocation algorithm, including transition functions, capacity-constrained redistribution, and movement inertia constraints, is provided in Appendix.

1) ANNUAL PILGRIM TOTALS

Annual pilgrim counts were obtained from the Saudi General Authority for Statistics (GASTAT). These totals define the population to be distributed across sites each hour during the Hajj period:

- 2021: 58,745 pilgrims (COVID-19 restricted)
- 2022: 899,353 pilgrims
- 2023: 1,845,045 pilgrims
- 2024: 1,833,164 pilgrims
- 2025: 1,673,230 pilgrims

The large interannual variation reflects the transition from pandemic restrictions to full-capacity pilgrimage and provides a diverse range of crowd densities for model evaluation.

2) RITUAL SCHEDULE AND SITE ELIGIBILITY

The Hajj pilgrimage follows a well-defined sequence of rituals across four principal locations: Haram, Mina, Arafat, and Muzdalifah. We define a day index $d \in \{-2, -1, 0, 1, 2, 3, 4, 5\}$ where $d = 0$ corresponds to the arrival in Mina (8 Dhul-Hijjah) and $d = 1$ corresponds to the Day of Arafat (9 Dhul-Hijjah). Subsequent days correspond to Eid al-Adha and the Tashreeq period.

Site occupancy follows the ritual schedule:

- **Haram:** Accessible throughout the pilgrimage with crowd levels modulated by daily prayer times.
- **Mina:** Primary residence for pilgrims during days 0 and 2–5.

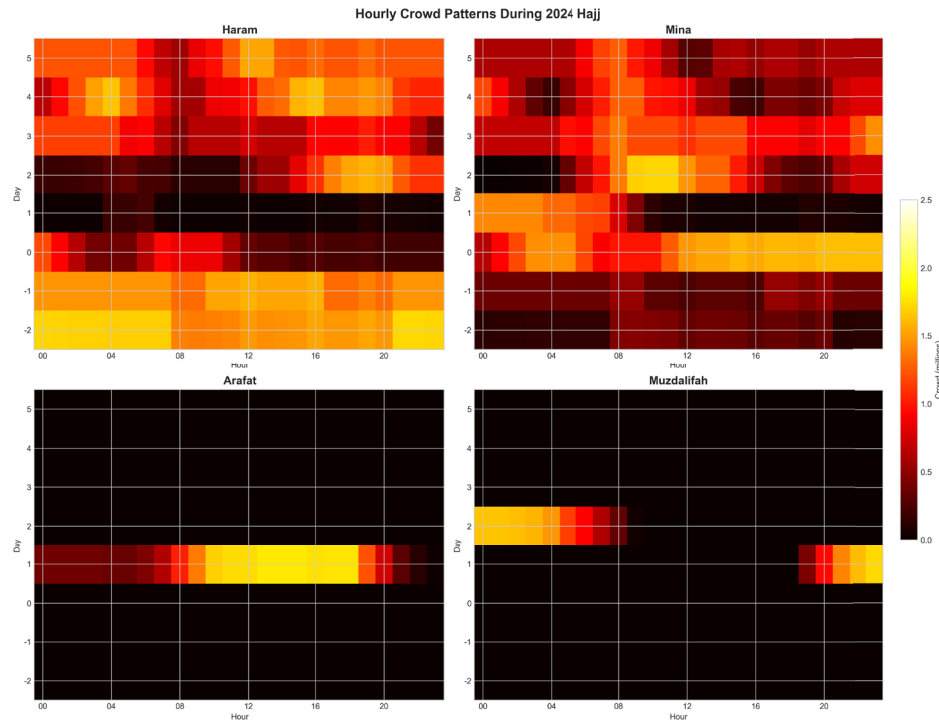


FIGURE 2. Hourly crowd patterns for the four Hajj sites during the 2024 pilgrimage. Each panel shows a heatmap of estimated crowd density (in millions) as a function of Hajj day (rows) and hour of day (columns). The patterns reflect site-specific ritual schedules: Haram experiences continuous occupancy with prayer-time surges; Mina hosts pilgrims during days 0–5 with overnight stays and stoning peaks; Arafat concentrates activity on day 1 between sunrise and sunset; Muzdalifah shows night-time occupancy on days 1–2 with peak near Fajr. These patterns are anchored to astronomical events (sunrise, sunset, Fajr, and prayer times) obtained from the AlAdhan API for the 2024 Hajj dates.

- **Arafat:** Occupied only during the Day of Arafat between sunrise and sunset.
- **Muzdalifah:** Occupied overnight following the departure from Arafat.

These rules determine which sites are eligible to host pilgrims during each stage of the pilgrimage.

3) ASTRONOMICAL ANCHORING

Transitions between ritual stages occur at times determined by solar and lunar cycles rather than fixed clock hours. To account for this variability, key transition events were anchored to astronomical prayer times obtained from the AlAdhan API using the Umm al-Qura calculation method.

The following astronomical events govern major transitions:

- Sunrise on the Day of Arafat: beginning of Mina → Arafat movement.
- Sunset (Maghrib) on the Day of Arafat: departure from Arafat toward Muzdalifah.
- Fajr on Eid morning: peak departure from Muzdalifah to Mina.
- Sunrise during the Tashreeq days: earliest permitted time for stoning rituals.

These astronomical anchors allow the model to reproduce year-specific timing shifts in the pilgrimage schedule.

4) DERIVED CROWD METRICS

The allocation model generates hourly crowd estimates for each site. From these allocations we derive several metrics used by the heat-risk model:

- **Estimated crowd:** number of pilgrims present at each site-hour.
- **Crowd density:** pilgrims per square kilometer.
- **Capacity utilization:** fraction of site capacity occupied.
- **Congestion level:** categorical indicator (LOW, MODERATE, HIGH, EXTREME).

The final dataset contains 3,840 hourly site-level records (5 years × 8 Hajj days × 24 hours × 4 sites). These records provide the crowd-density input $\rho_n(x, t)$ used in the hybrid heat-risk model.

Figure 2 illustrates the resulting crowd distribution patterns during the 2024 Hajj season. The estimated occupancy profiles reproduce the expected ritual dynamics, including daytime concentration at Arafat, nighttime occupancy at Muzdalifah, and sustained presence at Mina and Haram.

D. INTEGRATED DATASET STRUCTURE

The proposed framework integrates three complementary data sources representing atmospheric forcing, spatial susceptibility, and event-driven modifiers:

- 1) **Meteorological time series.** Hourly station observations of air temperature T_a , dew-point temperature T_d , and wind speed W for the Hajj seasons 2021–2025. These variables are used to compute relative humidity and wet-bulb temperature T_w , which defines the baseline physiological heat stress $H_0(t)$.
- 2) **Spatial morphological features.** A 1 km grid covering the Makkah region containing satellite-derived variables describing urban morphology, including Local Climate Zone (LCZ) class, building density, vegetation index (NDVI), and land-surface temperature diurnal range (ΔLST). These features are used to estimate the spatial amplification potential $A(x)$.
- 3) **Crowd intensity estimates.** Hourly estimates of pilgrim distribution across the main Hajj sites derived from the astronomy-aware crowd allocation model. From these allocations we derive a normalized crowd intensity variable $\rho_n(x, t)$ that represents the relative occupancy of each site.

All datasets are aligned temporally using hourly timestamps and spatially using the common 1 km analysis grid. Meteorological forcing is applied uniformly across the domain, while urban amplification is computed at grid-cell level and crowd intensity is mapped to the corresponding pilgrimage sites. The resulting integrated dataset supports both the machine-learning estimation of spatial amplification $A(x)$ and the physics-based computation of the heat-risk field $H(x, t)$.

E. EXPERIMENTAL PROTOCOL

The experimental protocol evaluates both the machine-learning component used to estimate spatial amplification and the overall performance of the hybrid heat-risk framework.

1) AMPLIFICATION MODEL EVALUATION

The machine-learning model predicting climatological surface excess $SE(x)$ from satellite-derived morphological features is evaluated using a temporal hold-out strategy. Observations from the years 2021–2024 are used for training, while 2025 is reserved as an independent test set.

Model performance is quantified using standard regression metrics, including mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). In addition, permutation feature importance is computed to verify that the learned relationships are consistent with physical expectations (e.g., positive influence of urban density and negative influence of vegetation).

2) HYBRID MODEL EVALUATION

After estimating the spatial amplification field $A(x)$, the full heat-risk model is evaluated by computing hourly heat stress fields $H(x, t)$ across all Hajj sites and seasons. The resulting effective wet-bulb temperature estimates are analyzed to assess the contribution of urban morphology, crowd intensity, and wind conditions to overall heat exposure.

In addition, robustness to parameter uncertainty is evaluated through a one-at-a-time sensitivity analysis, where key parameters are perturbed within a ± 20 –30% range and the resulting variations in effective heat stress and alert-level distributions are examined.

3) BASELINE COMPARISONS

To quantify the contribution of each component, the hybrid framework is compared with progressively simplified baselines:

- **Meteorological baseline:** $H(x, t) = T_w(t)$.
- **Meteorology + crowd + wind:** $H(x, t) = T_w(t) + \Delta T_{crowd}(x, t) + \Delta T_{wind}(t)$.
- **Full hybrid model:** $H(x, t) = T_w(t) + \Delta T_{urban}(x, t) + \Delta T_{crowd}(x, t) + \Delta T_{wind}(t)$.

These comparisons isolate the contribution of spatial urban amplification relative to purely atmospheric or crowd-driven components.

4) ALERT-LEVEL ASSESSMENT

To evaluate operational relevance, predicted effective wet-bulb temperatures are mapped to heat-risk alert levels (GREEN, YELLOW, ORANGE, RED, and BLACK). The analysis focuses on the temporal distribution, persistence, and site-specific frequency of these alert levels across years.

IV. SPATIAL AMPLIFICATION ESTIMATION VIA MACHINE LEARNING

The urban amplification potential $A(x)$ introduced in the framework represents the persistent spatial susceptibility of each location to elevate thermal stress relative to the regional meteorological baseline. Estimating this quantity directly from physical principles would require detailed information on three-dimensional building geometry, material thermal properties, and radiative exchange processes, which are rarely available at city scale.

To overcome this limitation, we estimate $A(x)$ from remotely sensed morphological features using machine learning. Importantly, machine learning is used only to infer this spatial susceptibility parameter, while temporal dynamics and thermophysiological transformations remain governed by explicit physical formulations within the hybrid framework.

The physical rationale for using surface excess as a learning target is as follows. Urban surfaces that store and re-radiate heat elevate near-surface air temperature, which in turn raises wet-bulb temperature through the psychrometric relationship. While humidity plays an independent role in determining wet-bulb temperature, the baseline meteorological component $H_0(t)$ already captures the full temperature–humidity coupling via the Stull formulation. The urban amplification term ΔT_{urban} therefore represents the additional air-temperature elevation caused by urban morphology, which contributes additively to effective wet-bulb temperature. Urban modification of local humidity

(e.g., through reduced evapotranspiration or anthropogenic moisture sources) is not explicitly modeled in the current framework and represents a direction for future refinement.

This separation preserves interpretability while allowing the model to exploit spatial information contained in satellite observations. Consequently, the proposed approach avoids the use of fully closed-box prediction models, ensuring that each component of the heat-risk estimate corresponds to a physically interpretable process. In this sense, machine learning acts as a data-driven parameterization of urban heat amplification within an otherwise physically structured heat-risk model.

It should be noted that the machine learning component itself (gradient boosting regression) does not impose physical constraints such as energy balance, monotonicity, or conservation laws during training. The physics-structured character of the overall framework resides in the additive decomposition into physically interpretable components, the physiologically grounded choice of wet-bulb temperature as the baseline metric, and the bounded parameter ranges informed by physical reasoning.

A. TARGET DEFINITION

To learn the spatial amplification potential, we require a measurable quantity that isolates the persistent thermal influence of urban morphology. We therefore define the *climatological surface excess* as the long-term mean difference between land surface temperature and ambient air temperature:

$$SE(x) = LST_{mean}(x) - T_{ERA5,mean}(x)$$

where $LST_{mean}(x)$ represents the multi-year mean land surface temperature derived from satellite observations (combining Terra and Aqua platforms), and $T_{ERA5,mean}(x)$ denotes the corresponding mean near-surface air temperature from the ERA5-Land reanalysis.

This difference captures the persistent excess temperature associated with local surface properties, including urban materials, vegetation cover, and thermal inertia. By averaging over multiple years, the resulting signal suppresses short-term weather variability and reveals the stationary spatial imprint of urban morphology.

Learning to predict $SE(x)$ from morphological features therefore allows the model to infer the spatial amplification potential $A(x)$ used in the urban heat-risk formulation.

B. FEATURE SET

For each 1 km grid cell in the study domain, we compile a set of satellite-derived features that characterize the urban surface environment:

- **Local Climate Zone (LCZ) class:** a categorical descriptor of urban form capturing typical building geometry, density, and surface materials.
- **Building density:** the fraction of land covered by building footprints, representing development intensity.

- **Normalized Difference Vegetation Index (NDVI):** a spectral indicator of vegetation cover that influences surface cooling through evapotranspiration and shading.
- **LST diurnal range (ΔLST):** the difference between Aqua (afternoon) and Terra (morning) land surface temperatures, serving as a proxy for surface thermal inertia.

These predictors were selected based on physical reasoning: LCZ and building density capture urban morphology, NDVI represents vegetation cooling, and ΔLST reflects surface heat storage capacity. Together, they provide a compact representation of the main factors governing spatial heat amplification.

C. LEARNING MODEL

The mapping from morphological features to surface excess is learned using a Histogram-based Gradient Boosting Regressor. This algorithm is well suited for geospatial regression tasks because it can capture nonlinear relationships, accommodate mixed data types (categorical LCZ classes and continuous predictors), and provide robust predictive performance on moderate-sized datasets.

The learned model can be expressed as

$$\widehat{SE}(x) = f_{ML}(LCZ(x), BD(x), NDVI(x), \Delta LST(x))$$

where f_{ML} denotes the trained regression model and the predictors represent satellite-derived morphological characteristics of each grid cell.

Model training follows a temporal hold-out validation strategy:

- **Training set:** grid-cell observations derived from years 2021–2024.
- **Test set:** observations from 2025, reserved for independent evaluation of generalization performance.

Although the morphological predictors are largely time-invariant, the target variable $SE(x)$ is derived from multi-year satellite observations and therefore reflects interannual variability in surface thermal conditions. Using a temporal hold-out ensures that model evaluation reflects generalization to unseen years rather than memorization of year-specific spatial patterns.

Model performance is assessed using standard regression metrics, including mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). These metrics quantify the ability of the model to reproduce spatial variations in climatological surface excess across the study region.

D. NORMALIZATION TO AMPLIFICATION POTENTIAL

The predicted surface excess $\widehat{SE}(x)$ represents the estimated difference between land surface temperature and atmospheric temperature in degrees Celsius. To convert this quantity into the dimensionless spatial susceptibility index used in the heat-risk model, predictions are first normalized using robust

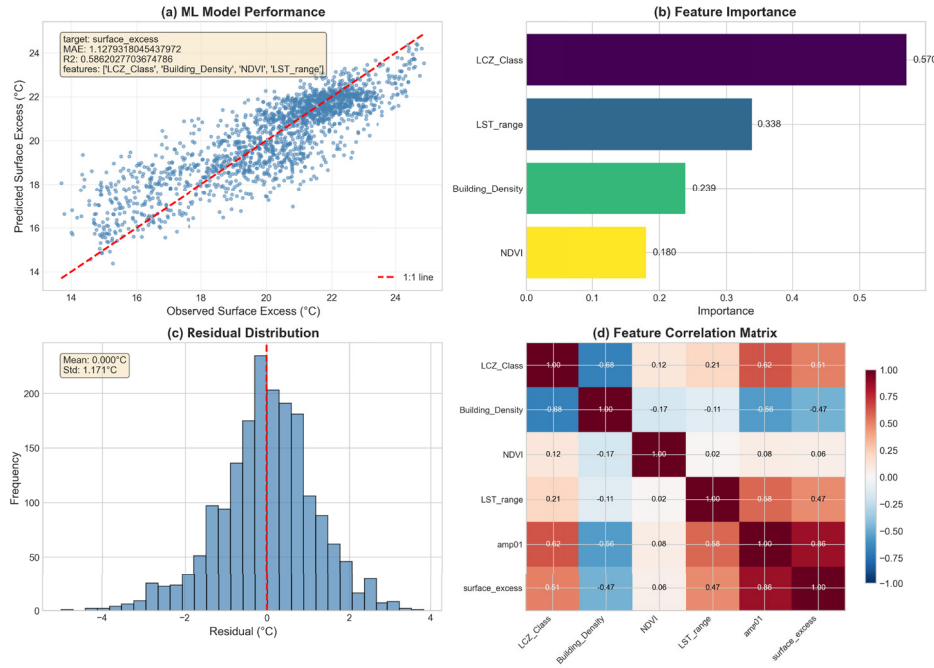


FIGURE 3. Amplification model performance: (a) predicted versus observed surface excess; (b) permutation feature importance; (c) residual distribution; (d) feature correlation matrix.

percentiles:

$$A_{raw}(x) = \frac{\widehat{SE}(x) - P_5}{P_{95} - P_5}$$

where P_5 and P_{95} denote the 5th and 95th percentiles of predicted surface excess across all grid cells. This transformation reduces the influence of extreme outliers while preserving the relative spatial structure of amplification.

The normalized index is then clipped to the unit interval and scaled to the physically bounded amplification range:

$$A(x) = \Delta T_{max} \cdot \min(1, \max(0, A_{raw}(x)))$$

where $\Delta T_{max} = 5.0^\circ\text{C}$ represents a conservative upper bound on urban heat amplification, consistent with reported ranges of urban heat island intensity in the literature [30].

The resulting $A(x)$ field quantifies the spatial susceptibility of each location to urban heat amplification and is subsequently used within the urban amplification term defined in Section II-C.

V. RESULTS

The proposed hybrid framework was applied to the 2021–2025 Hajj seasons, generating hourly heat-risk estimates for the four principal pilgrimage sites: Haram, Mina, Arafat, and Muzdalifah. The results are organized in six parts: (i) evaluation of the machine-learning model used to estimate spatial amplification, (ii) spatial structure of the learned urban amplification field, (iii) decomposition of effective heat stress into its additive physical components, (iv) alert-level distributions during the extreme 2024 Hajj season, (v) cross-year comparison across the 2021–2025 study period, and

TABLE 3. Amplification model performance for surface excess prediction.

Metric	Value
Target Variable	$SE(x)(LST_{mean} - T_{ERA5,mean})$
Number of Samples	1,855
Mean Absolute Error (MAE)	1.13 °C
Root Mean Squared Error (RMSE)	1.17 °C
Coefficient of Determination (R^2)	0.586
Permutation Feature Importance	
LCZ Class	0.570
LST Diurnal Range (ΔLST)	0.338
Building Density	0.239
NDVI	0.180

(vi) parameter sensitivity analysis assessing the robustness of the framework to uncertainty in key model parameters.

A. AMPLIFICATION MODEL PERFORMANCE

The Histogram-based Gradient Boosting Regressor used to estimate climatological surface excess $SE(x)$ was evaluated on 1,855 grid cells after excluding cells with incomplete feature coverage. Table 3 summarizes the predictive performance and permutation feature importance.

The amplification model explains 58.6% of the variance in climatological surface excess, with an MAE of 1.13 °C. Given that the model relies on only four satellite-derived predictors and that surface excess reflects the combined effects of urban form, material properties, vegetation, and background climate, this level of performance indicates that

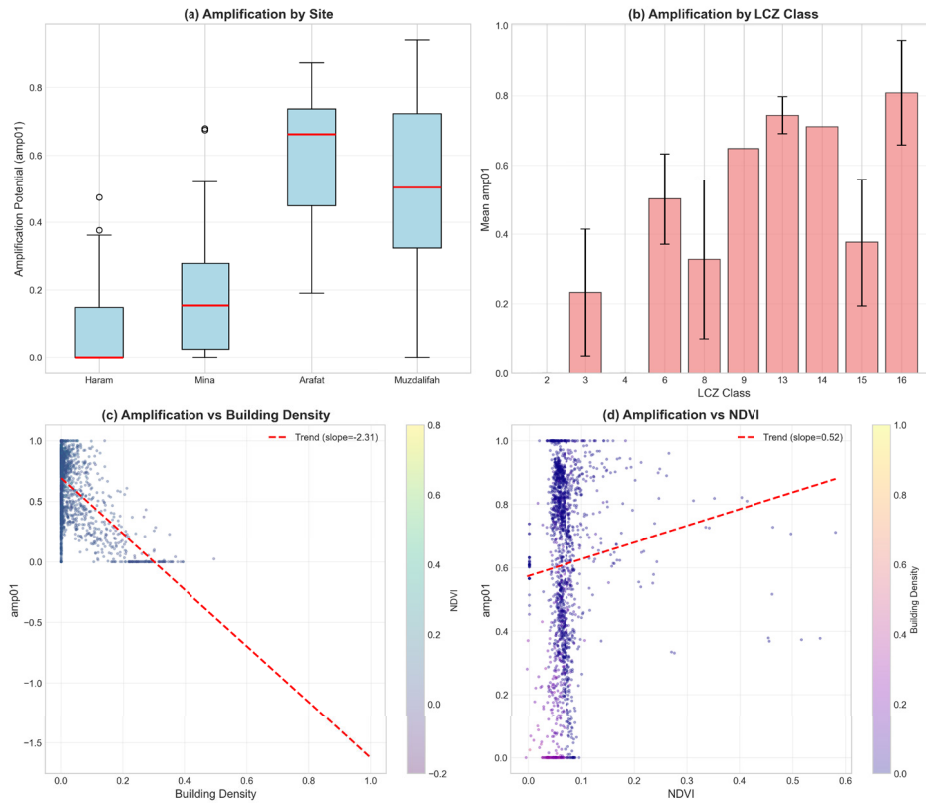


FIGURE 4. Spatial patterns of amplification potential: (a) site-level distribution of *amp01*; (b) mean *amp01* by LCZ class; (c) relationship with building density; (d) relationship with NDVI.

the selected features capture a substantial portion of the persistent spatial signal governing urban heat amplification.

The remaining unexplained variance likely reflects the influence of micro-scale urban characteristics, transient atmospheric conditions, surface material heterogeneity, and measurement uncertainties that are not captured by the available predictors.

The ordering of feature importance is also physically informative. LCZ class is the dominant predictor, indicating that broad urban-form typologies provide the strongest signal for distinguishing thermally amplified and weakly amplified surfaces. LST diurnal range is the second most important variable, which is consistent with its role as a proxy for thermal inertia and surface heat storage capacity. Building density and NDVI make smaller but still meaningful contributions, capturing development intensity and vegetation cooling effects, respectively. This pattern supports the interpretation that persistent urban amplification in the study area is driven primarily by built-form type and heat-storage behavior rather than by vegetation alone.

Figure 3 provides a more detailed view of model behavior. The predicted-versus-observed scatterplot shows that the model reproduces the broad structure of the target variable without evidence of severe bias, while the residual distribution is centered close to zero and remains comparatively narrow. The feature-correlation panel further indicates that

TABLE 4. Spatial amplification potential (*amp01*) by site.

Site	Mean	Std	Min	Max	N
Arafat	0.603	0.177	0.190	0.873	47
Haram	0.082	0.136	0.000	0.474	28
Mina	0.182	0.182	0.000	0.680	45
Muzdalifah	0.507	0.235	0.000	0.940	54

the learned amplification field is not simply reproducing any single raw predictor, but rather combining multiple morphological signals into a coherent spatial estimate.

Overall, the amplification model performs well enough to support its intended role within the hybrid framework: not as a closed-box predictor of final heat risk, but as a data-driven estimator of the persistent spatial susceptibility field $A(x)$ that enters the physically structured heat-risk model.

B. SPATIAL PATTERNS OF URBAN AMPLIFICATION

The machine-learned amplification field exhibits pronounced spatial heterogeneity across the four Hajj sites. Table 4 summarizes the site-level distribution of the normalized amplification index *amp01*, where *amp01* denotes the normalized form of $A(x)$ scaled to the unit interval $[0, 1]$. For clarity, $A(x)$ refers to the amplification field used in the model formulation, whereas *amp01* denotes its

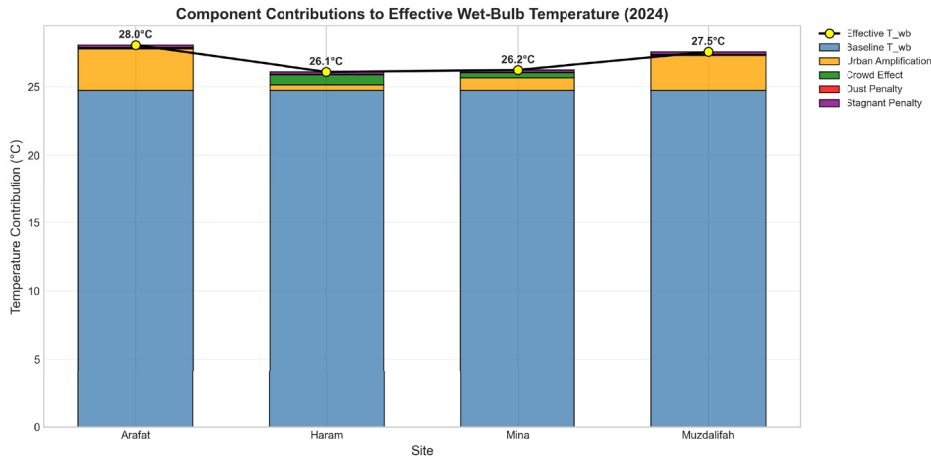


FIGURE 5. Component contributions to effective wet-bulb temperature during the 2024 Hajj season. Urban amplification dominates at Arafat and Muzdalifah, while crowd contributions are largest at Haram and Mina.

TABLE 5. Mean component contributions to effective wet-bulb temperature during the 2024 Hajj.

Site	Baseline T _{wb}	Urban Amp.	Crowd	Dust	Stagnant	Effective T _{wb}
Arafat	24.73 °C	3.02 °C	0.08 °C	0.02 °C	0.19 °C	28.04 °C
Haram	24.73 °C	0.41 °C	0.73 °C	0.02 °C	0.23 °C	26.12 °C
Mina	24.73 °C	0.91 °C	0.37 °C	0.02 °C	0.21 °C	26.25 °C
Muzdalifah	24.73 °C	2.54 °C	0.07 °C	0.02 °C	0.19 °C	27.55 °C

normalized representation used for visualization and site-level comparison. Figure 4 visualizes this variability across sites and morphological contexts.

Arafat has the highest mean amplification potential, followed by Muzdalifah, whereas Haram shows consistently low amplification and Mina occupies an intermediate position. This ranking is important because it indicates that the four sites are not equally susceptible to heat storage and re-radiation, even before crowd and wind effects are considered. Instead, the pilgrimage network contains a clear hierarchy of inherent thermal susceptibility, with Arafat and Muzdalifah standing out as the most amplified environments.

The contrast between Arafat and Haram is particularly informative. Arafat combines a high mean amplification (0.603) with a relatively narrow spread, indicating that elevated thermal susceptibility is spatially widespread across the site rather than confined to a few localized hotspots. Haram, by contrast, has a very low mean amplification (0.082), confirming that its surface characteristics and urban structure contribute much less to persistent heat amplification. Mina exhibits moderate mean amplification but substantial within-site variability, suggesting a more heterogeneous morphology in which some zones behave more like thermally amplified urban surfaces while others remain closer to the regional baseline. Muzdalifah combines high mean amplification with the largest variability, consistent with a site where strong heat-storage zones coexist with less amplified open areas.

Figure 4 reinforces these site-level patterns. The LCZ-based breakdown shows that the learned amplification

field is sensitive to morphological classes, while the relationships with building density and NDVI indicate that the model is capturing thermal contrasts shaped jointly by built form, exposed surfaces, and limited vegetation. Although the raw relationships are not perfectly monotonic, the overall pattern confirms that the learned *amp01* field encodes more than a simple density gradient: it reflects a broader thermal landscape governed by urban structure and surface behavior.

From an operational perspective, these results indicate that urban morphology preconditions the spatial risk landscape. Sites such as Arafat and Muzdalifah begin from a higher thermal baseline once urban amplification is added, which helps explain why these locations later emerge as the most severe hotspots in the full heat-risk model.

C. COMPONENT CONTRIBUTIONS TO EFFECTIVE HEAT STRESS

To identify the dominant mechanisms driving site-level heat exposure, effective wet-bulb temperature was decomposed into baseline meteorological forcing, urban amplification, crowd contribution, dust-related penalty, and stagnant-air penalty. Table 5 reports the mean contribution of each component during the 2024 Hajj season, and Figure 5 visualizes their relative magnitudes.

Because the baseline wet-bulb forcing is common across sites, all spatial differences in effective heat stress arise from the additive modifiers. This makes the decomposition particularly informative: it highlights which sites are most

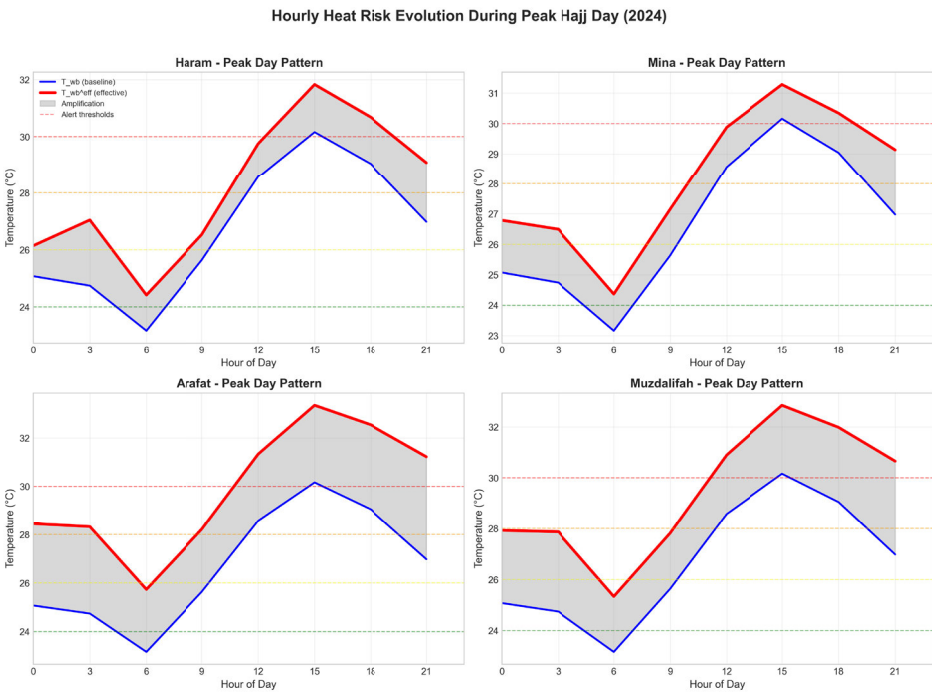


TABLE 6. Hourly alert frequency by site during the 2024 Hajj.

Site	GREEN	YELLOW	ORANGE	RED	BLACK
Arafat	7 (11.1%)	8 (12.7%)	12 (19.0%)	19 (30.2%)	17 (27.0%)
Haram	14 (22.2%)	15 (23.8%)	17 (27.0%)	14 (22.2%)	3 (4.8%)
Mina	14 (22.2%)	11 (17.5%)	18 (28.6%)	18 (28.6%)	2 (3.2%)
Muzdalifah	6 (9.5%)	10 (15.9%)	18 (28.6%)	16 (25.4%)	13 (20.6%)

exposed and indicates the dominant contributing factors under the modeled conditions.

Arafat and Muzdalifah are dominated by urban amplification, with mean urban contributions of 3.02 °C and 2.54 °C, respectively. These are large increments relative to the common meteorological baseline and help explain why both sites approach or exceed the highest alert thresholds despite only weak crowd contributions. In other words, the results suggest that the dominant contributor to extreme heat risk at these two sites is the thermal behavior of the built and exposed surface environment rather than local crowd intensity.

Under the modeled crowd distributions, Haram exhibits the opposite regime. Its urban contribution is minimal (0.41 °C), but the crowd contribution reaches 0.73 °C, the highest among all sites. This suggests that Haram is not intrinsically the most thermally amplified site from a morphological perspective; instead, its risk is strongly shaped by crowd-modified microclimate. Mina represents an intermediate case: urban amplification is non-negligible (0.91 °C), but crowd contribution remains substantial (0.37 °C),

indicating a mixed risk regime driven by both built form and occupancy.

Wind-related penalties remain comparatively small in absolute magnitude, yet they are systematically present. Stagnant-air effects exceed dust-related effects at every site, indicating that poor ventilation is a more consistent source of additional thermal burden than high-wind dust entrainment during the observed Hajj conditions.

Taken together, these results suggest four distinct site-level risk regimes: urban-dominated risk at Arafat, nocturnally amplified urban-dominated risk at Muzdalifah, crowd-dominated risk at Haram, and mixed urban–crowd risk at Mina. This is one of the most practically important findings of the study, because it implies that mitigation strategies should be site-specific rather than uniform.

D. ALERT-LEVEL DISTRIBUTION DURING THE 2024 HAJJ
The color-coded alert system translates effective wet-bulb temperature into five operational categories: GREEN, YELLOW, ORANGE, RED, and BLACK. Table 6 summarizes the

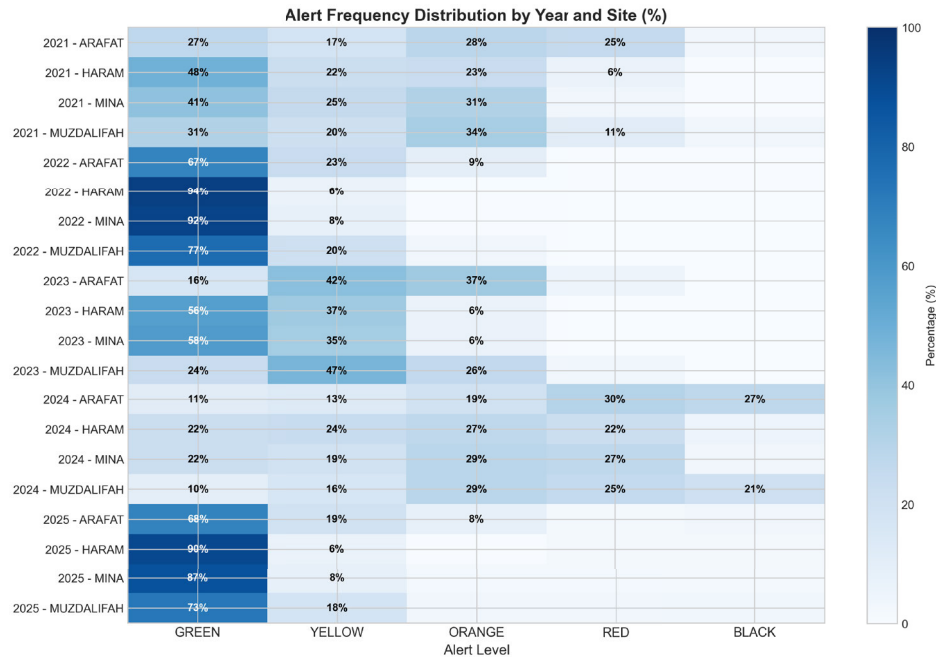


FIGURE 7. Alert frequency heatmap across all sites and years (2021–2025). The 2024 Hajj season is the only year with substantial BLACK-alert frequency, especially at Arafat and Muzdalifah.

hourly alert frequencies across the four Hajj sites during the 2024 season.

The 2024 Hajj season is characterized by severe and persistent high-risk conditions, particularly at Arafat and Muzdalifah. Arafat records the highest proportion of BLACK-alert hours (27.0%), followed by Muzdalifah (20.6%). These elevated frequencies are consistent with the component-level analysis, which showed that both sites are strongly dominated by urban amplification. The addition of 2.5 °C to 3.0 °C of urban-induced heat load systematically shifts effective wet-bulb temperatures toward critical thresholds, resulting in frequent transitions into the most dangerous alert category.

The distribution of alert levels also reveals the persistence of high-risk conditions. At Arafat, RED and BLACK alerts together account for more than 57% of all hours, indicating that the site remains in severely stressful conditions for extended periods rather than experiencing only short-lived extremes. Muzdalifah exhibits a similar pattern, with a large fraction of hours in ORANGE, RED, and BLACK categories. In contrast, Haram and Mina show more balanced distributions, with lower BLACK-alert frequencies (4.8% and 3.2%, respectively) and a greater proportion of hours in moderate alert levels.

These differences reflect the distinct physical mechanisms governing heat risk at each site. Arafat and Muzdalifah are primarily influenced by urban and surface-related amplification, which elevates the entire temperature distribution. Haram and Mina, by contrast, are more strongly influenced by crowd density, which produces localized but less persistent increases in thermal stress.

Figure 6 provides further insight into the temporal structure of these alert conditions. On the peak day of 2024, Arafat exceeds the BLACK-alert threshold during the afternoon, coinciding with maximum solar exposure and atmospheric heat load. Muzdalifah, however, exhibits sustained elevated conditions into the evening and nighttime period, reflecting delayed heat release from the surface environment. This temporal asymmetry is important from a health perspective, as it reduces the opportunity for physiological recovery between consecutive days of exposure.

Figure 7 extends this analysis across all sites and years. The contrast is clear: 2024 is the only season with substantial BLACK-alert frequency, and this burden is concentrated primarily at Arafat and Muzdalifah. Other years exhibit lower and more localized high-risk conditions, indicating that the extreme outcomes observed in 2024 arise from the combined effect of elevated baseline meteorological conditions and site-specific amplification.

Overall, the alert-level analysis confirms that the 2024 Hajj season is not only extreme in terms of peak conditions but also in terms of the duration and spatial extent of high-risk exposure. The results further demonstrate that the distribution of alert levels is strongly shaped by site-specific physical drivers, reinforcing the need for differentiated heat-risk mitigation strategies across the pilgrimage network.

Importantly, the identification of 2024 as the most severe heat-risk season is temporally consistent with observed real-world outcomes. The 2024 Hajj was associated with more than 1,300 reported heat-related fatalities under extreme ambient temperatures exceeding 51.8 °C. The elevated

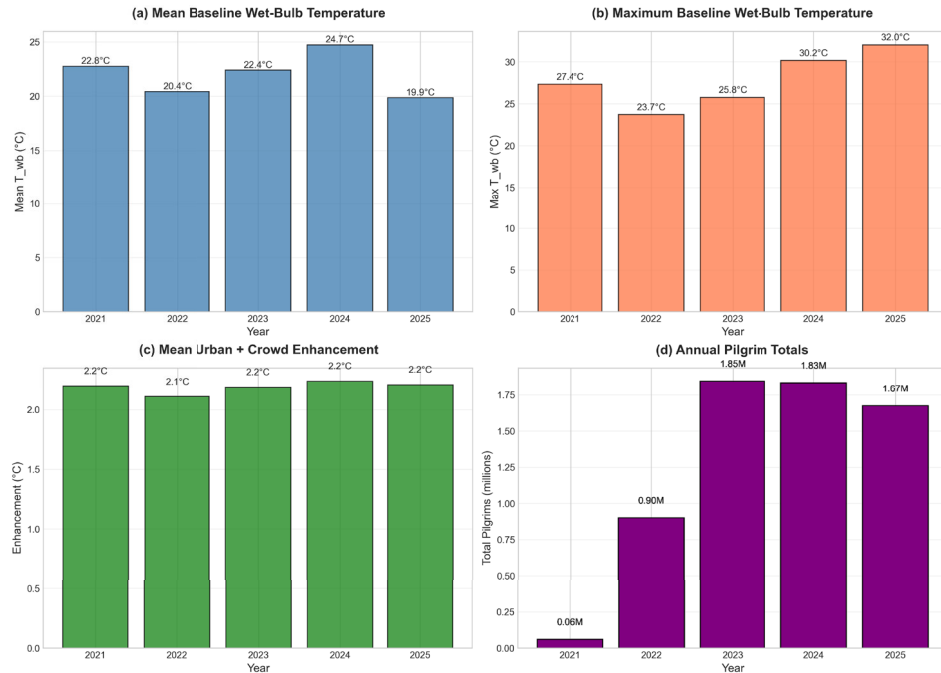


FIGURE 8. Cross-year comparison of key metrics. The 2024 season exhibits the highest mean wet-bulb temperature among full-capacity Hajj years and the largest sustained effective heat load.

TABLE 7. Cross-year comparison of heat-risk metrics (2021–2025).

Year	Mean T_w	Max T_w	Mean T_w^{eff}	Max T_w^{eff}	Enhancement
2021	22.8°C	27.4°C	25.0°C	30.6°C	2.2°C
2022	20.4°C	23.7°C	22.6°C	27.3°C	2.1°C
2023	22.4°C	25.8°C	24.6°C	29.1°C	2.2°C
2024	24.7°C	30.2°C	27.0°C	33.4°C	2.3°C
2025	19.9°C	32.0°C	22.1°C	36.4°C	2.2°C

frequency and persistence of BLACK-alert conditions predicted by the model during this period align with the timing of the documented escalation in heat-related health impacts. However, this agreement should not be interpreted as validation of the model’s spatial or population-specific predictions. Notably, more than 80% of reported fatalities occurred among unregistered pilgrims walking long distances without adequate shelter [7], representing exposure pathways not explicitly modeled within the present site-based framework, which focuses on the structured ritual locations. The correspondence between the modeled seasonal severity and the observed health outcomes therefore provides evidence of qualitative temporal consistency rather than quantitative validation.

E. CROSS-YEAR COMPARISON

To place the 2024 season in context, Table 7 compares baseline and effective heat-risk metrics across the five-year study period.

Among the five years considered, 2024 has the highest mean baseline wet-bulb temperature, the highest mean

effective heat stress, and the largest sustained high-risk burden across the Hajj sites. This result is important because it distinguishes 2024 not merely as a season with isolated extremes, but as the year with the most persistently elevated physiological heat stress.

The comparison also shows why maximum values alone are insufficient to characterize seasonal severity. Although 2025 records a higher maximum effective wet-bulb temperature than 2024, its mean baseline and mean effective values are substantially lower. In other words, 2025 contains more isolated extremes, whereas 2024 combines high peaks with sustained elevation over longer periods. This distinction is fully consistent with the alert analysis, which identifies 2024 as the only year with widespread BLACK-alert burden.

Figure 8 summarizes these interannual differences. The 2024 season stands apart not because its urban or crowd enhancement is radically larger than in other years, but because a comparable enhancement is superimposed on the warmest baseline meteorological conditions. This shows that extreme risk emerges from the interaction between atmospheric forcing and local amplification rather than from either component alone.

F. PARAMETER SENSITIVITY ANALYSIS

To assess the robustness of the proposed framework to parameter uncertainty, a one-at-a-time sensitivity analysis was conducted by perturbing key model parameters within a ± 20 –30% range. The parameters tested include the maximum urban amplification ΔT_{max} , the crowd contribution

TABLE 8. Summary of parameter sensitivity analysis.

Parameter	Baseline	Range Tested	ΔT_{wb}^{eff} (mean)	BLACK Alert Change
Urban amplification (ΔT_{max})	5.0 °C	±30%	± 0.5 °C	1.75% – 5.40%
Crowd contribution (κ)	1.2 °C	±30%	0.1 °C to 0.2 °C	3.49% – 3.73%
Stagnant-air penalty	0.8 °C	±30%	< 0.1 °C	minor
Dust penalty	0.8 °C	±30%	< 0.1 °C	negligible
Diurnal modulation (γ)	0.18	±30%	negligible	none

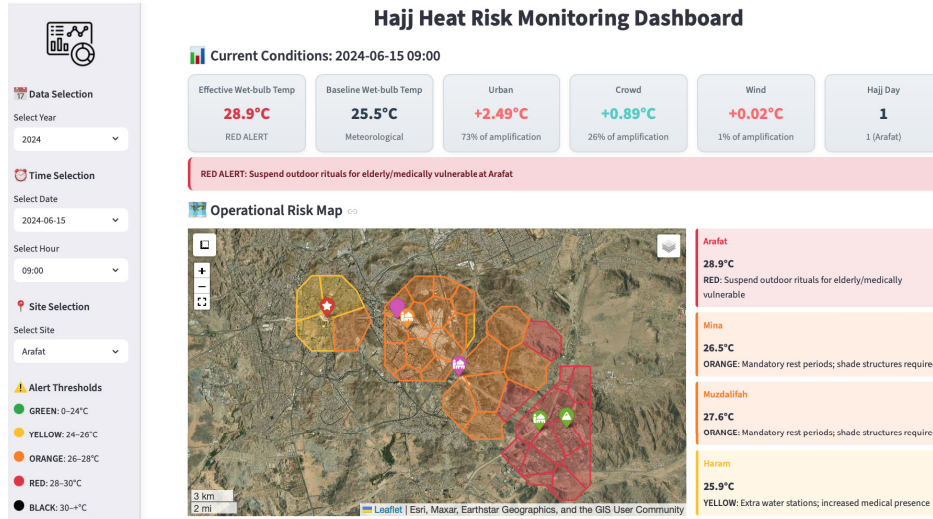


FIGURE 9. Operational Hajj heat monitoring dashboard showing spatial heat-risk distribution across the four pilgrimage sites together with the additive decomposition of effective wet-bulb temperature and the five-level alert classification system used for operational monitoring.

coefficient κ , and the wind-related penalty terms, as well as the amplitude of the diurnal modulation function.

The results indicate that the model is most sensitive to the urban amplification parameter. Variations of ΔT_{max} produce changes of approximately ± 0.5 °C in mean effective wet-bulb temperature and substantial variation in the frequency of extreme (BLACK) alert conditions. This confirms that urban heat storage is the dominant driver of extreme heat risk in the modeled system.

In contrast, variations in the crowd contribution parameter κ lead to more moderate changes, with mean effective temperature varying by approximately 0.1–0.2 °C and only limited impact on alert frequencies. Wind-related parameters have a comparatively minor effect, influencing mean effective temperature by less than 0.1 °C. Similarly, the diurnal modulation parameter does not affect aggregated temperature statistics but modifies the temporal distribution of heat exposure.

Importantly, across all tested parameter configurations, the relative ranking of sites remains unchanged, with Arafat and Muzdalifah consistently exhibiting the highest heat-risk levels, while Haram and Mina remain comparatively less amplified. This indicates that the main conclusions of the study are structurally robust and not driven by specific parameter choices.

A quantitative summary of the sensitivity analysis is provided in Table 8.

VI. OPERATIONAL DECISION-SUPPORT SYSTEM

Beyond retrospective analysis, the hybrid heat-risk framework has been implemented as a prototype decision-support platform designed to demonstrate the feasibility of integrating the different modeling components into a real-time heat-risk monitoring environment. The system is intended to illustrate how spatial heat-risk information could support scenario-based analysis and operational planning during mass gatherings rather than to provide fully validated operational guidance. A public prototype implementation of the system is accessible at:

<https://hajjheatmonitor.dpdns.org>

It integrates the hybrid modeling framework with real-time meteorological observations, site-level crowd estimates, and spatial amplification fields derived from satellite data. Together, these components enable the generation of continuously updated estimates of effective wet-bulb temperature and heat-risk alerts across the four principal Hajj sites: Haram, Mina, Arafat, and Muzdalifah.

The operational platform translates the framework into an interactive monitoring environment for scenario-based

TABLE 9. Alert-level thresholds and operational interpretation.

Level	Color	T_{wb}^{eff} Range	Operational Interpretation
1	GREEN	$< 24^{\circ}\text{C}$	Minimal thermophysiological strain
2	YELLOW	24°C to 26°C	Caution advised; risk increases during prolonged exertion
3	ORANGE	26°C to 28°C	Elevated heat stress; cooling and hydration strongly recommended
4	RED	28°C to 32°C	Dangerous conditions with high risk of heat illness
5	BLACK	$\geq 32^{\circ}\text{C}$	Extreme heat stress; outdoor activity suspension or severe restriction advised

analysis and decision support during the Hajj season and comparable mass gatherings.

A. DASHBOARD INTERFACE

The platform provides an interactive dashboard that visualizes both current and forecasted heat-risk conditions across the pilgrimage network. The central indicator is the effective wet-bulb temperature $H(x, t)$, displayed together with its additive physical decomposition into:

- baseline atmospheric wet-bulb temperature,
- urban amplification,
- crowd metabolic contribution,
- wind-related modulation.

As illustrated in Figure 9, this decomposition enables users to identify not only where thermal risk is elevated but also which physical mechanism is responsible for the increase. For example, risk peaks at Arafat and Muzdalifah are typically driven by strong urban amplification, whereas Haram frequently exhibits higher crowd-related contributions during peak ritual periods.

This interpretability is a key advantage of the hybrid framework, allowing operational users to understand the causal drivers of heat stress rather than observing a single aggregated risk index.

B. SPATIAL RISK VISUALIZATION

Spatial heat-risk maps are generated using Voronoi tessellation of the underlying 1 km grid cells used in the spatial amplification model. Each polygon represents the area of influence of a grid cell and is colored according to the current alert level derived from the effective wet-bulb temperature.

This representation produces an intuitive operational zoning system that can support the identification of localized heat hotspots and inform potential resource allocation strategies. Examples include the allocation of medical teams, installation of temporary cooling stations, or adjustment of crowd circulation routes in areas experiencing elevated thermal stress.

C. ALERT SYSTEM

The platform implements a five-level color-coded alert system based on effective wet-bulb temperature thresholds, summarized in Table 9. The thresholds are informed by the established understanding of thermophysiological strain under humid heat conditions and are intended as operational heat-risk categories rather than strict clinical diagnostic boundaries. Wet-bulb temperatures in the range

28°C to 32°C correspond to conditions under which exertional heat illness risk rises sharply, particularly for vulnerable populations and individuals engaged in sustained physical activity [10], [12]. Sustained wet-bulb temperatures above 32°C approach conditions associated with severe uncompensable heat stress even under limited physical exertion. The BLACK threshold ($\geq 32^{\circ}\text{C}$) therefore reflects conditions where outdoor activity should be suspended or extremely limited, broadly consistent with extreme heat-health warning practices and occupational heat-stress guidance frameworks internationally.

The thresholds were selected to prioritize operational interpretability and conservative heat-risk communication under extreme mass-gathering conditions. Although the proposed thresholds are expressed in terms of effective wet-bulb temperature rather than WBGT, they are broadly consistent with established occupational heat-stress guidance frameworks based on physiological heat exposure limits [33], [34].

Alert levels are updated continuously and summarized through time-series plots and statistical distributions, enabling operators to track both short-term fluctuations and broader temporal patterns. Because the alert system is derived from the additive decomposition of $H(x, t)$, it remains directly interpretable in terms of the underlying physical processes driving heat risk.

By operationalizing the physics-guided hybrid framework, the system demonstrates the feasibility of real-time, spatially explicit, and mechanism-aware heat-risk monitoring, providing a practical foundation for decision-support in high-density mass gatherings under extreme climatic conditions.

VII. DISCUSSION

The results demonstrate that heat risk during the Hajj pilgrimage does not arise from atmospheric conditions alone, but from the interaction between meteorological forcing, urban morphology, and crowd-induced modifications. The proposed hybrid framework makes this interaction explicit and quantifiable, providing insight into the mechanisms that govern extreme heat exposure in mass-gathering environments.

A. MECHANISTIC INTERPRETATION OF HEAT-RISK DRIVERS

A central contribution of this work is the decomposition of effective wet-bulb temperature into physically interpretable

components. This reveals that different sites are governed by distinct dominant mechanisms.

At Arafat and Muzdalifah, urban amplification is the primary driver, contributing more than 2.5 °C on average. This reflects persistent effects of surface properties, thermal inertia, and solar exposure, which elevate the entire temperature distribution rather than producing isolated extremes.

In contrast, Haram and Mina are more strongly influenced by crowd density under the modeled crowd distribution scenarios, where localized metabolic heat release and reduced ventilation act as dominant modifiers. These effects are more spatially concentrated and temporally variable, particularly during peak ritual periods.

This contrast defines two distinct heat-risk regimes: (i) *morphology-driven* (Arafat, Muzdalifah), where risk is structurally elevated, and (ii) *crowd-driven* (Haram, Mina), where risk is dynamically modulated. This distinction has direct operational implications, as it suggests that mitigation strategies must be site-specific.

B. INTERPRETATION OF THE 2024 EXTREME EVENT

The 2024 Hajj season provides a critical example of compound heat risk. While baseline wet-bulb temperatures were elevated, the severity of the event cannot be explained by meteorology alone.

Instead, extreme conditions result from the combination of:

- elevated baseline wet-bulb temperature,
- strong urban amplification at Arafat and Muzdalifah,
- and sustained exposure during peak ritual periods.

The additive structure shows that moderate increases in baseline conditions can push already vulnerable sites beyond critical thresholds. This explains the marked increase in RED and BLACK alert hours in 2024.

Temporal patterns further differentiate sites: Arafat experiences peak afternoon exposure, while Muzdalifah exhibits prolonged evening and nighttime heat due to delayed release of stored energy. This persistence limits physiological recovery and likely contributes to cumulative heat stress over consecutive days.

The model-predicted seasonal severity ranking identifies 2024 as the most extreme Hajj year within the study period. This aligns temporally with the documented increase in heat-related fatalities during the 2024 Hajj and provides qualitative evidence of consistency between the modeled heat-risk field and real-world heat-stress outcomes at the event scale. The concentration of sustained RED and BLACK alert conditions at Arafat and Muzdalifah corresponds to locations and time periods where heat exposure was likely most severe. However, the model operates at the site level and does not explicitly represent the exposure pathways of unregistered pilgrims, who constituted the majority of reported victims. This correspondence therefore suggests that the framework captures plausible large-scale heat-risk dynamics during extreme Hajj conditions, but it should not

be interpreted as validation of site-specific or population-specific heat-risk predictions.

C. CROSS-YEAR STABILITY AND STRUCTURAL ROBUSTNESS

Despite interannual variability in meteorological conditions and pilgrim numbers, the structure of heat-risk drivers remains stable across 2021–2025. The spatial ranking of sites by amplification is consistent, indicating that urban morphology acts as a persistent modifier of heat risk.

Similarly, the magnitude of combined effects produces a systematic increase of approximately 2.1 °C to 2.3 °C relative to baseline wet-bulb temperature. This consistency suggests that the model captures structural system properties rather than year-specific variability.

In addition to cross-year consistency, the robustness of the framework to parameter uncertainty was evaluated through a complementary sensitivity analysis (Section V-F). The results indicate that urban amplification is the dominant source of sensitivity, with variations of approximately $\pm 0.5^\circ\text{C}$ in effective wet-bulb temperature, while crowd and wind parameters have more moderate effects. Importantly, the relative ranking of sites and the identification of dominant heat-risk regimes remain stable across all tested parameter configurations, reinforcing the structural robustness of the model.

D. LIMITATIONS AND UNCERTAINTY CONSIDERATIONS

While the results reveal consistent structural patterns, several sources of uncertainty should be considered.

First, the crowd distribution is derived from a synthetic but physically constrained model rather than direct measurements. Although it captures key dynamics of pilgrim movement through ritual schedules, annual pilgrim totals, astronomical anchoring, and physical movement constraints, it has not been validated against empirical crowd-count observations such as gate sensors, mobile phone records, or camera-based pedestrian density estimates. Consequently, the model does not capture real-time variability or localized congestion, and the resulting ΔT_{crowd} estimates should be interpreted as scenario-based approximations contingent on the assumed crowd distributions rather than measured occupancy conditions. Future integration of empirical crowd-sensing datasets, subject to availability and privacy considerations, would substantially improve the reliability and realism of this component.

Second, key model parameters including $\Delta T_{\text{max}} = 5.0^\circ\text{C}$, $\kappa = 1.2^\circ\text{C}$, and the wind-modulation penalties are adopted from general urban-climate and crowd-physiology literature rather than calibrated specifically for Makkah's extreme aridity and unique crowd-density conditions. As demonstrated in the sensitivity analysis (Section V-F), variations in ΔT_{max} alone can substantially alter the frequency of BLACK-alert conditions. Consequently, absolute numerical outputs such as precise alert frequencies and effective wet-bulb temperature values should be interpreted with appropriate caution.

Nevertheless, the relative ranking of sites by heat-risk severity and the identification of dominant risk regimes (urban-dominated versus crowd-dominated) remain structurally stable across the tested parameter ranges. Site-specific calibration using dedicated field measurements during future Hajj or non-Hajj periods would substantially improve confidence in the absolute magnitude of the predicted heat-risk levels.

Third, the additive formulation assumes independence between components, whereas real-world interactions may be nonlinear. For example, crowd presence can simultaneously modify local humidity and airflow, thereby altering the effectiveness of evaporative cooling—effects that are not explicitly represented by linearly additive terms. The present formulation should therefore be interpreted as a first-order approximation designed to capture the dominant independent contributions of each mechanism while neglecting higher-order interaction effects. Quantifying the error introduced by this simplification would require either high-resolution computational fluid dynamics simulations or controlled in-situ measurements capable of resolving coupled crowd-microclimate dynamics, both of which are beyond the scope of the current study. Despite this limitation, the additive decomposition provides physically interpretable insight into the primary drivers of spatial heat-risk variability across the Hajj sites.

Fourth, validation is limited by the lack of spatially resolved in-situ heat-stress measurements. The proposed framework should therefore be interpreted as a physically structured, scenario-based heat-risk estimation approach rather than a fully validated predictive microclimate model. While the spatial amplification component was quantitatively evaluated against climatological surface excess derived from satellite observations and reanalysis data, the final effective wet-bulb temperature field $H(x, t)$ could not be directly validated against dense in-situ measurements because such observations are not currently available across the Hajj sites. Consequently, the resulting heat-risk estimates should be interpreted as physically plausible first-order approximations intended for comparative spatial analysis and decision-support exploration rather than precise operational forecasts.

Fifth, the use of surface excess $SE(x)$ as a learning target for spatial amplification assumes that urban-induced air-temperature enhancement represents the dominant pathway linking surface properties to effective wet-bulb amplification. Urban influences on local humidity are not explicitly represented, which may lead to underestimation of amplification effects in areas where surface characteristics also modify moisture availability or evaporative conditions.

Finally, meteorological inputs from only two stations (Mina and Arafat) are extrapolated across four sites with distinct local microclimatic characteristics. In particular, the Haram area, located within a dense urban canyon environment, and Muzdalifah, characterized by more open terrain, likely experience different local temperature,

humidity, and wind conditions than those measured at the station locations. This spatial extrapolation therefore introduces additional uncertainty into the site-specific heat-risk estimates, particularly for the wind-modulation component.

Despite these limitations, the consistency of results across multiple years suggests that the framework captures the dominant first-order drivers of heat risk at a comparative spatial scale. Accordingly, the framework is primarily intended for comparative spatial heat-risk assessment and exploratory decision-support analysis rather than deterministic operational forecasting.

E. IMPLICATIONS FOR HEAT-RISK MANAGEMENT IN MASS GATHERINGS

Building on these findings, the results highlight the need for mechanism-specific heat-risk mitigation strategies. For morphology-driven sites such as Arafat and Muzdalifah, interventions should focus on shading, surface materials, and passive cooling. For crowd-driven environments such as Haram and Mina, strategies should prioritize density control, ventilation, and crowd flow optimization.

More broadly, the findings demonstrate that physiological heat-risk metrics must be combined with spatial modeling. Approaches based solely on air or surface temperature cannot capture the compound effects observed in dense mass gatherings.

As extreme heat events become more frequent under climate change, compound risk environments of this type are likely to intensify. The ability to distinguish between atmospheric forcing and local amplification mechanisms is therefore critical. In this context, the proposed framework provides a generalizable and interpretable approach for supporting heat-risk assessment and targeted intervention in complex, high-density environments.

VIII. OPEN-SOURCE IMPLEMENTATION AND REPRODUCIBILITY

To promote transparency and reproducibility, the modeling framework and processed datasets used in this study are made publicly available through the following open-source repository:

<https://github.com/fhamdime/crowd-wetbulb-heat-risk>

The repository includes the complete data-processing pipeline, the physics-guided hybrid modeling framework, the spatial amplification learning component, the crowd estimation modules, and the scripts used to reproduce all results and figures presented in this paper. It also includes the processed datasets, feature-construction pipeline, model-training scripts, and reproducibility notebooks used in the study.

The repository is organized to facilitate reproducibility of the experimental workflow, including data preparation, model training, evaluation, and figure generation.

IX. CONCLUSION AND FUTURE DIRECTIONS

This paper introduced a physics-guided hybrid framework for spatial modeling of crowd-modified wet-bulb heat risk during the Hajj pilgrimage. The approach combines a physiologically grounded meteorological baseline with additive components representing urban amplification, crowd-induced heat load, and wind-related modulation, while using machine learning to estimate spatial susceptibility from satellite-derived morphological features.

The results show that heat risk in mass gatherings emerges from the interaction of multiple mechanisms rather than from atmospheric conditions alone. Distinct site-specific regimes are identified: Arafat and Muzdalifah are dominated by persistent urban amplification, whereas Haram and Mina are more strongly influenced by crowd density under the modeled scenarios. This mechanistic differentiation explains why heat exposure varies substantially across sites despite identical meteorological forcing.

Across the 2021–2025 period, the framework exhibits stable and interpretable behavior. The spatial amplification structure remains consistent, and the combined effect of local modifiers produces a systematic increase of approximately 2.1 °C to 2.3 °C in effective wet-bulb temperature relative to the baseline. The 2024 Hajj season stands out as the most severe case, where elevated baseline conditions coincide with strong spatial amplification, leading to sustained high-risk exposure.

Methodologically, this work demonstrates how machine learning can be integrated within a physically structured framework to infer spatial heterogeneity while preserving interpretability. From an operational perspective, the model supports site-specific mitigation strategies by distinguishing between morphology-driven and crowd-driven heat-risk regimes, and its deployment as a real-time decision-support system illustrates its practical applicability.

Future work should focus on integrating short-range meteorological forecasts for predictive risk assessment, improving crowd modeling through real-time sensing data, and enriching spatial descriptors with higher-resolution urban morphology and material properties.

A critical priority is the deployment of portable micro-climate sensors and wearable heat-stress monitors at key pilgrimage sites to enable direct comparison between modeled and observed effective wet-bulb conditions, thereby establishing end-to-end validation of the framework. Dedicated field campaigns to calibrate key model parameters, including ΔT_{max} , κ , and wind-modulation coefficients, specifically for Makkah's arid environment and the Hajj crowd context, would substantially improve confidence in absolute heat-risk estimates and are recommended before operational reliance on the system. Similarly, deployment of dedicated meteorological stations at the Haram and Muzdalifah sites would enable site-specific atmospheric forcing and help reduce spatial extrapolation uncertainty.

Future model refinements could also explore multiplicative or physically constrained nonlinear coupling terms between urban amplification, crowd-induced humidity, and wind modulation, subject to availability of higher-resolution validation data. Coupling the framework with human thermophysiological models would further enable transition from environmental hazard estimation to population-specific risk assessment.

Beyond the Hajj context, the overall modeling framework—namely the decomposition of heat risk into baseline atmospheric forcing, spatial urban amplification, and event-specific crowd modifiers—is transferable in its general structural principles to other large-scale mass gatherings. However, practical implementation would require substantial contextual adaptation. In particular, the spatial amplification model would need to be retrained using local urban morphological features, while the crowd-allocation component would need to be replaced with event-appropriate mobility or crowd-estimation approaches. The Hajj-specific ritual schedules and astronomical anchoring used in the present study are therefore not directly transferable to other events. Application of the framework to different mass-gathering environments would thus preserve the physics-structured decomposition strategy while adapting the site-specific and event-specific modeling components.

In summary, this study provides a robust, interpretable, and operationally relevant framework for understanding and managing heat risk in complex, high-density environments.

APPENDIX CROWD ALLOCATION MODEL

This appendix provides a detailed description of the synthetic crowd allocation model used to generate hourly pilgrim distributions across the four principal Hajj sites (Haram, Mina, Arafat, and Muzdalifah). The model produces physically plausible crowd dynamics consistent with the ritual schedule of the Hajj pilgrimage and the physical constraints of pedestrian movement between sites.

The allocation procedure integrates four sources of information:

- 1) Annual pilgrim totals obtained from official statistics,
- 2) Ritual schedules governing site eligibility,
- 3) Astronomical transition times determining key ritual events,
- 4) Physical constraints on crowd movement and site capacities.

Together, these components enable the generation of hourly site-level crowd distributions that reproduce the expected temporal structure of the pilgrimage.

A. RITUAL ACTIVITY PROFILES

For each site s and pilgrimage day d , we define a baseline activity profile $f_{\pi(s,d)}(h)$ describing the expected relative intensity of occupancy across hours of the day h .

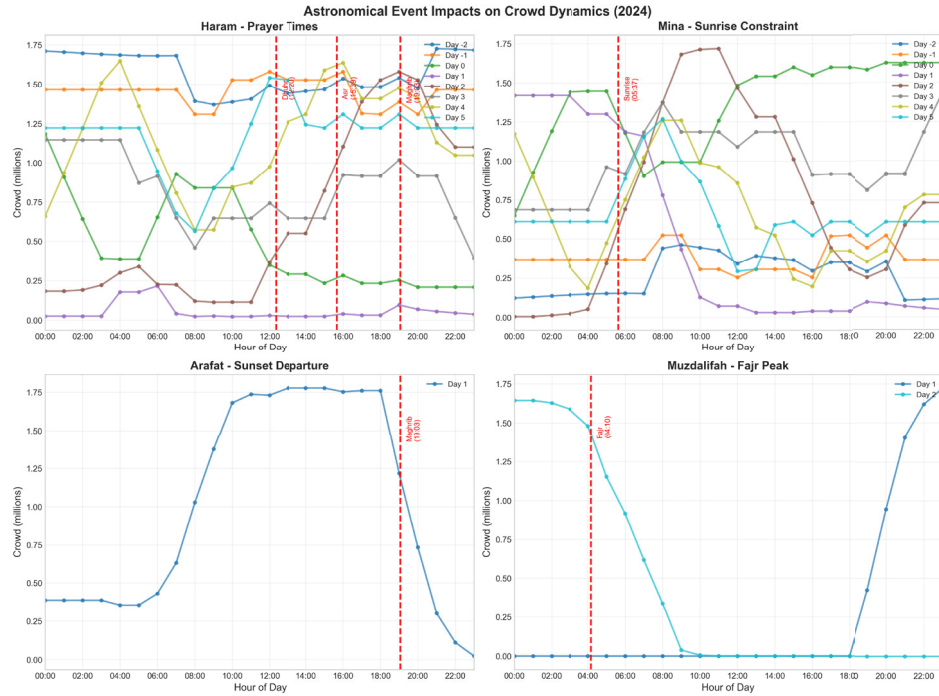


FIGURE 10. Impact of astronomical events on crowd dynamics during the 2024 Hajj. Each panel shows the peak-day hourly crowd profile for a site, with vertical lines indicating key prayer times and ritual transition moments derived from the AIAdhan API using Umm al-Qura calculation method. The alignment between crowd surges and astronomical events validates the model's astronomy-aware anchoring.

These profiles capture the temporal structure of the Hajj rituals, including overnight camping in Mina, the daytime standing ritual at Arafat, and the overnight stay in Muzdalifah. The profiles represent relative activity intensities rather than probabilities and are later normalized during the allocation procedure.

B. ASTRONOMICAL EVENT ANCHORING

Key transitions between pilgrimage sites are determined by astronomical events rather than fixed clock hours. To account for this variability, we anchor transitions to prayer times derived from the AIAdhan API using the Umm al-Qura calculation method.

The principal astronomical anchors include:

- Sunrise on the Day of Arafat (beginning of Mina → Arafat movement),
- Sunset (Maghrib) on the Day of Arafat (departure from Arafat toward Muzdalifah),
- Fajr on Eid morning (peak movement from Muzdalifah to Mina),
- Sunrise during the Tashreeq days (earliest permitted stoning time).

Anchoring transitions to these events ensures that the allocation model adapts to the year-specific timing of the pilgrimage.

C. TRANSITION FUNCTIONS

To avoid unrealistic discontinuities in site occupancy, transitions between sites are implemented using smooth ramp functions anchored to astronomical events.

Three principal transition mechanisms are modeled:

- **Mina → Arafat transition:** a linear ramp over four hours following sunrise on the Day of Arafat.
- **Arafat → Muzdalifah transition:** a linear ramp beginning after sunset and lasting approximately five hours.
- **Muzdalifah → Mina transition:** a logistic ramp centered around the Fajr prayer.

These ramp functions gradually redistribute pilgrims between sites while maintaining temporal continuity.

D. CAPACITY-CONSTRAINED ALLOCATION

For each hour (y, d, h) we compute raw allocation scores for each site based on activity profiles, eligibility constraints, and transition ramps. These scores are normalized to produce site weights

$$w_{y,d,h}(s)$$

such that

$$\sum_s w_{y,d,h}(s) = 1.$$

The desired allocation is then computed as

$$\tilde{x}_{y,d,h}(s) = N_y \cdot w_{y,d,h}(s)$$

where N_y represents the total number of pilgrims in year y .

If the desired allocation exceeds the physical capacity C_s of a site, an iterative redistribution procedure reallocates excess pilgrims among remaining eligible sites while preserving the total population.

E. MOVEMENT INERTIA CONSTRAINTS

To prevent unrealistic hour-to-hour changes in site occupancy, movement inertia constraints are imposed:

$$|x_h(s) - x_{h-1}(s)| \leq L_h(s)$$

where $L_h(s)$ denotes the maximum allowable change in occupancy for site s between consecutive hours.

These limits reflect physical constraints on pedestrian movement between pilgrimage sites. After enforcing these constraints, allocations are renormalized to preserve the total number of pilgrims.

F. ILLUSTRATION OF ASTRONOMICAL EVENT ALIGNMENT

Figure 10 illustrates how the estimated crowd profiles align with key astronomical events during the 2024 Hajj season. Vertical lines correspond to sunrise, sunset, and prayer-time anchors obtained from the AlAdhan API.

The alignment between crowd surges and astronomical anchors confirms that the allocation model reproduces the temporal structure of the Hajj ritual schedule while maintaining physically plausible transitions.

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FAYÇAL HAMDI received the M.Sc. and Ph.D. degrees in computer science from the University of Paris-Saclay, France, in 2008 and 2011, respectively. He is currently an Associate Professor with the Faculty of Computer and Information Systems, Islamic University of Madinah. He has been actively involved in several national and European research projects, including H2020 projects, such as SAFECARE and Polifonia; and ANR and CNRS initiatives in linked data, knowledge engineering, and data quality. He has collaborated with both academic and industrial partners on applications related to cultural heritage, cybersecurity, and smart data systems. He has supervised several Ph.D. theses in the fields of semantic web, data quality, and knowledge-based systems. His current research explores hybrid approaches combining machine learning and domain knowledge for modeling complex systems and supporting real-world decision-making. His research interests include artificial intelligence, semantic web technologies, ontology engineering and alignment, data integration, and large-scale knowledge representation.



MOHAMED BENAÏDA received the B.Sc. degree in computer science and information systems and the Ph.D. degree in human-computer interaction (HCI) and usability from the University of Salford, U.K. He is a Professor with the Islamic University of Madinah, Saudi Arabia. He is recognized as a Researcher in HCI, usability engineering, web design, e-government systems, and emerging technologies, including blockchain and the IoT. His work emphasizes practical applications with attention to cultural considerations in technology design and real-world impact in governance and smart systems.



HACHEMI BENNACEUR received the master's degree from Pierre and Marie Curie University, Paris, and the Ph.D. degree from Sorbonne University, Paris-Nord Campus, France. He is a Professor of computer science with Imam Mohammad Ibn Saud Islamic University. From 1992 to 2009, he held academic positions in France, serving as an Associate Professor with Cergy-Pontoise University and later as a Full Professor with Artois University, before joining his current institution. During this period, he was affiliated with the Laboratoire d'Informatique de Paris Nord (LIPN), Sorbonne Paris Nord and the Computer Science Research Center of Lens (CRIL). His research interests include several areas of artificial intelligence, including constraint programming, metaheuristics and optimization, and machine learning and its applications. He has supervised several master's and Ph.D. theses in these areas.



YASSER ALDWYHAN received the M.Sc. and Ph.D. degrees in computer science (cloud computing) from the University of Melbourne, Australia. He is currently an Assistant Professor with the Faculty of Computer and Information Systems, Islamic University of Madinah, Saudi Arabia. Prior to this, he was a Researcher with Saudi National Center for Electronics, Communications, and Photonics, King Abdulaziz City for Science and Technology (KACST), one of the leading research institutions in Saudi Arabia. His research interests include cloud, fog, edge computing, the Internet of Things (IoT), and smart city applications.



HANI ALMOAMARI received the Ph.D. degree in computer science from the University of Stirling. He is currently an Associate Professor with the Faculty of Computer and Information Systems, Islamic University of Madinah, and the Vice Dean for Quality Assurance. He is the co-author of many articles in software engineering, e-learning, networking, blockchain, security, and privacy. His current research interests include network security, privacy, and the IoT.



ABDULAZIZ ALSHAHRANI received the B.Sc. degree in information technology from King Abdulaziz University, Saudi Arabia, the M.Sc. degree in information technology from La Trobe University, Australia, and the Ph.D. degree in human-centered computing from the University of Warwick, U.K. He is currently an Associate Professor with the Faculty of Computer and Information Systems, Islamic University of Madinah, where he is also the Head of the Information Technology Department. He has an extensive experience in digital transformation and emerging technologies. He has been actively involved as a Consultant and an Expert in digital government initiatives. He collaborates with international organizations on global assessments of digital government performance, including online services, infrastructure, and human capital development. His research interests include artificial intelligence, digital government, human-centered computing, data science, and emerging technologies, such as the IoT and blockchain. His work focuses on bridging technology, policy, and strategic decision-making to support large-scale digital innovation and governance.



ABDULLAH ALSHAMQITI received the B.Sc. degree in computer science from Taibah University, Madinah, Saudi Arabia, and the M.Sc. and Ph.D. degrees from the University of Leicester, U.K. In 2012, he joined the Faculty of Computer Science and Information Technology, Islamic University of Madinah, as a Lecturer, and was appointed as an Assistant Professor of smart systems and software reverse engineering, in 2018. He has recognized for his work on software reverse engineering based on dynamic analysis and model/graph transformations using intelligent learning and inference approaches. He is interested in extending his knowledge by welcoming any research cooperation in different cutting edge disciplines, including quantum machine learning, hybrid AI approaches that focus on solving NLP challenges, and the interpretability of deep learning models using graph transformation rules.

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